CHAPTER 10

BUSINESS INTELLIGENCE SYSTEMS AND EIS

Quotation

Today the cutting edge technologies reside with consumers. From there, this technology filters into the enterprise. This fundamental shift means application vendors will have to give users what they want: Sophisticated applications in an easy-to-understand wrapper.

Don Campbell

Structure

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♣ QUICK LOOK AT CHAPTER THEMES

Over the years, data sources grow in number and size. Managers, supervisors and executive teams have obtained key information from many EIS applications. As per the wikipedia,[1] 2006, "Business intelligence entails the gathering of data from internal and external

[1] The Free Encyclopedia, http://en.wikipedia.org (accessed 17 July 2006).

data sources, as well as the storing and analysis thereof to make it accessible to assist in better decision-making." Business intelligence (BI) is arguably the most important enterprise application in an organization. BI is the technology and practice of applying information to make decisions. In this way, BI is different than data warehousing, which is more about storing information. It is also more comprehensive than data mining. Information shows its real value when many people can use and share it. This is the goal of business intelligence.[2] Figure 10.1 gives a brief idea about the BI with the help of a diagram.

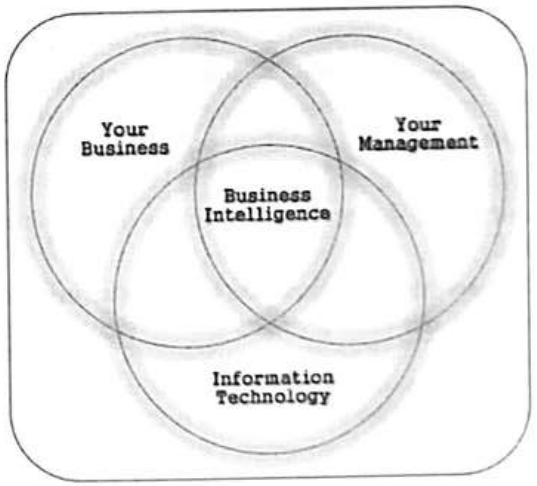


Figure 10.1: BI the Intersection of IT, Management, Business.

Source: sju-online[3]

[2] Cognos, an IBM company, is the world leader in business intelligence (BI) and performance management solutions. It provides world-class enterprise BI, planning and consolidation software and services to help companies plan, understand and manage financial and operational performance. Cognos brings together technology, analytical applications, best practices, and a broad network of partners to give customers a complete performance solution. It provides an open and adaptive solution that leverages an organization's ERP, packaged applications, and database investments.

[3] http://www.sju-online.com/programs/business-intelligence-masters.asp

♣ LEARNING OBJECTIVES

The objective of this chapter is to give students an understanding of various aspects of business intelligence systems and knowledge management with a managerial focus. Specifically, the areas include:

- ♣ Concepts to understand the role of business intelligence in the fields of marketing, finance, human resources, and production.
- ♣ How to gain fundamental understanding of Business Intelligence technologies.
- ♣ How to develop a decision-making approach to business intelligence technology.
- ♣ How to understand business intelligence framework and architecture.
- ♣ How to develop a foundation in Business Intelligence for Business Analysis.
- ♣ How to appreciate the role of Business Intelligence in the decision-making process of an organization.
- ♣ How to understand the key success factors in a business intelligence environment.
- ♣ How to assess and design a Business Intelligence infrastructure.
- ♣ How to describe and plan Business Intelligence operations and their management.

♣ CASE STUDY

National Scorecard Transforms How Best Buy Does Business

Best Buy[4] is North America's number-one specialty retailer of consumer electronics, personal computers, entertainment software and appliances. With over 700 stores nationwide and rapid growth continuing, Best Buy needed an application that was simple to use, delivered consistent information on which to run the business, and was intuitive enough to be used by everyone in the company.

SUMMARY

With disparate data sources and fragmented reports, a solution was needed to align 90,000 employees in making decisions based on timely, consistent information. The result was a national scorecard that now serves as a single source graphical representation of information that unifies priorities, tracking and reporting on key business areas.

ABOUT LANCET

Lancet has been able to retain top-notch employees due to our award-winning work environment. Our culture fosters open discussion, teamwork and creativity with a constant emphasis on training and growth. Many clients have found our style to be pleasantly contagious; they hear our feedback in the form of business advice and we encourage and welcome their feedback. Lancet has created a process for interaction and idea exchange among our employees, allowing us to continue learning as individuals and as a company. When you hire a Lancet employee, you get the benefit of the whole Lancet team, including formalized quality assurance, support and an internal knowledge base. Our employees are enthusiastic and not only thrive on learning and mastering new technologies, but also on sharing their knowledge with one another and our clients.

[4] Best Buy Co., Inc. is a specialty retailer of consumer electronics in the United States accounting for 19% of the market. Best Buy is sometimes called the "big blue box" because of the prominent design on Best Buy stores. One can go through the website at http://www.bestbuy.com/.

THE SITUATION

Historically, Best Buy had several disparate data sources with various users having access to relatively small sets of metrics producing fragmented reports while the demand for information and more complex ad-hoc reporting continued to increase exponentially. The data that would satisfy the demand was neither contained nor accessible within one data source, nor could employees access key information. And, out-of-the-box analysis software was not sophisticated enough to provide the ad-hoc drilling needed to make decisions based on root cause analysis. With no way to access and analyze the information essential to drive this fast-paced transaction-oriented business, Best Buy turned to Lancet for help.

THE SOLUTION

For seven years, Best Buy looked for a partner that was business savvy with IT vision and technical expertise to deliver an application that would feed critical information to every channel in the company, driving daily workflow, and do it all at a low cost of ownership. There was no centralized data repository, no clear path from key information to its root cause for analysis, no way to custom drill, nor a way to perform ad-hoc inquiries and get reports to retail without putting IT processes at risk. "Lancet was the only company we found that could realize our vision. They brought strategy, thought-leadership and business intelligence expertise," said Brad Elo, Director, Margin Enhancement Strategies. With years of business intelligence experience, Lancet worked with Best Buy's internal team to develop the technology strategy that delivered on Best Buy's vision. They also played a key role in executing the plan — mapping out logical connections and creating a custom functional, workflow-based drilling architecture and ad-hoc reporting system for the enterprise-wide national scorecard which has become a cornerstone application within Best Buy because of the way it has transformed the business. The resulting solution allows the business to get the information they need directly without going to IT, resulting in lower costs while maintaining the security and integrity of the production systems. In addition, the complexities and technical challenges are easily managed behind the scenes while employees are presented with a front-end that is simple to use and intuitive yet extremely powerful. "Lancet continues to play a key role in the project, including long term vision for the application," adds Elo.

THE RESULTS

Companywide, the national scorecard is the key to Best Buy's marketing and company intelligence. It is the single source that aligns and unifies the retail organization's priorities, tracking and reporting on key business areas, presenting information to users at all levels in the company in a format that is meaningful to them. The application touches every aspect of the business and keeps all 90,000 employees informed and focused on the same consistent information across the enterprise. Everyday these reports guide proactive behaviors essential to keeping Best Buy in the forefront of the fast-paced, consumer electronics retail business. Each week, over 1 million reports are generated that present accurate snapshots of the business from all angles. For the first time, employees at the stores can get useful customer information and understand what items or categories are "hot," answering key questions about what customers need and how sales associates can better interact in order to enhance the customer experience and foster passionate customer loyalty. Best Buy is more proactive and responds rapidly because trends can easily be identified. Problem areas or issues can be set to, "What If," scenarios, and access to data and response time is immediate. Plus, the data resources are so unified and rich that drilling anywhere is possible. Operations can see up-to-date labor allocation, usage, and spending. Sales performance metrics are readily accessible and time spent in front of the computer has been reduced, allowing store personnel more time with the customers. Retail behavior can be immediately adjusted and better paths to action are employed. With accurate, consistent information at their fingertips, employees know what to do and what not to do everyday. The return on investment is astonishing. For each $1 spent on the scorecard application, Best Buy yields a $4 return. Says Elo, "Lancet turned our vision into reality, transforming the way we do business — from the CEO to store management to customer specialists — with insight into information that helps us proactively manage our business while delivering a four-fold return on investment." The possibilities for the future of the national scorecard are endless and Lancet will play an integral part. "Our national scorecard has proved itself to be the most

strategic application in the company. It keeps all 90,000 employees focused on the key business drivers that give us the competitive advantage not possible before. Working with Lancet is a big win for Best Buy," concludes Elo.

Source: Lancet Software.[5]

♣ SKETCH OF BI RELEVANCE TO EIS

In today's rapidly moving and progressively more digital world, it is necessary to comprehend advancing technologies in order to accomplish anything. Business leaders need the capability to accumulate and digest information concerning customers, suppliers and competitors, and to formulate decisions that positively influence their company's performance. Enterprise systems offer powerful tools to better measure and control organizational operations. Many organizations have found that this valuable tool can be enhanced to provide even greater value through the addition of powerful business intelligence systems. Business intelligence guru Ralph Kimball defines business intelligence as "a generic term to describe leveraging the organization's information assets for making better business decisions."[6] Business intelligence in this context is supported by storing data (data warehouse and related systems) and conducting studies using this data to solve business problems (one means to do this is through data-mining). Figure 10.2 describes a view of how business intelligence can be built.

One of the most popular forms of data mining in EIS systems is support of customer relationship management (CRM). Data warehouses are one of the most popular extensions to EIS systems, with over two-thirds

[5] Lancet Software was founded in 1997 to provide business intelligence services. It has successfully completed hundreds of business intelligence and web development projects with its partners. © Copyright 2008.

[6] Kimball, R. and Ross, M. (2002). *The Data Warehouse Toolkit*, 2nd Ed. New York, NY: John Wiley and Sons.

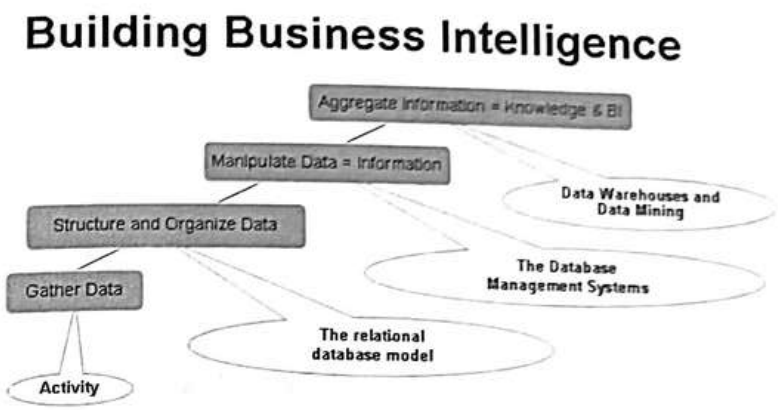


Figure 10.2: Building Business Intelligence.

Source: Exonous (2004).[7]

of U.S. manufacturers adopting or planning such systems (and slightly less than two-thirds of Swedish manufacturers doing the same).[8] This chapter discusses forms of data storage available. It also introduces data mining concepts, and reviews typical data mining applications within EIS systems.

♣ INTRODUCTION

Engaging in e-commerce is not a decision that should be made without first gaining knowledge about the firm and its relationships with its customers, competitors, suppliers, and other external entities. Business Intelligence (BI) is the activity of gathering information about the elements in the environment that interact with the firm. Business Intelligence endows you with a diverse business education in the field of technology and decision sciences and helps you to develop the capability to generate

[7] http://exonous.typepadc.com/mis/2004/03/building_busine.html

[8] Mabert, V. M., Soni, A. and Venkataramanan, M. A. (2000). Enterprise resource planning survey of US manufacturing firms. *Production and Inventory Management Journal*, 41(20), 52–58; Olhager J. and Selldin, E. (2003). Enterprise resource planning survey of Swedish manufacturing firms. *European Journal of Operational Research*, 146, 365–373.

business models for forecasting and business analysis, a fundamental understanding of Business Intelligence technologies, and the skills to integrate information from all aspects of the organization in understanding the big picture. Many organizations possess data in their databases, and generate additional valuable data. Business Intelligence (BI) is the process of collecting and turning this resource into business value.

Data is a key source of intelligence and competitive advantage for business organizations. With the explosion of electronic data available to organizations and the demand for better and faster decisions, the role of data-driven intelligence is becoming central in organizations. Data mining is the process of converting the raw data into useful knowledge required to support decision-making. It automates the process of knowledge discovery, providing us with greatly enhanced productivity in our search for useful information than we would be otherwise. It also increases the confidence with which we can make business decisions. In this age of the so-called information democracy, each employee of a company has the right to appropriate access to information. This alone (and there are many other reasons) turns the entire company into a potential business intelligence user base. This chapter will examine the value of information in a competitive environment from the perspectives of various types of business information, cost and management of information, developments on the Internet, and the role of governments. In addition, discerning client needs, and the packaging of information for client use are considered. The chapter will begin with the definition of business information, and work through specific types of information including company research, industry research and understanding statistics. The latter half of the chapter will focus on giving students an overview of the important trends in business research: competitive intelligence, knowledge management, and learning how to make a business case to support the decision to implement expensive information products. Virtually every business organization these days is in the process of exploring and implementing data mining solutions to core business problems. This chapter is essential for anyone interested in understanding how to get the maximum value from data, especially when abundant data are available. Application areas covered include marketing, customer relationship management, financial forecasting, risk management, personalization, Web searching, and other topics.

10.1 ALL ABOUT BI

There are a number of definitions of Business intelligence:

- BI as a term that encompasses a broad range of analytical software and solutions for gathering, consolidating, analyzing and providing access to information in a way that is supposed to let an enterprise's users make better business decisions.[9]
- BI benefits that facilitate the connections in the new-form organization, bringing real-time information to centralized repositories and support analytics that can be exploited at every horizontal and vertical level within and outside the firm.[10]
- BI includes an effective data warehouse and also a reactive component capable of monitoring the time-critical operational processes to allow tactical and operational decision-makers to tune their actions according to the company strategy.[11]
- BI as the result of in-depth analysis of detailed business data, including database and application technologies, as well as analysis practices.[12]
- BI as a technically much broader tool, including potentially knowledge management, ERP, decision support systems and data mining.[13]

[9] Adelman, S., Moss, L. and Barbusinski, L. (2002). I found several definitions of BI. *DM Review*, www.dmreview.com/article_sub.cfm?articleId (accessed 10 Sep 2008).

[10] Malhotra, Y. (2000). From information management to knowledge management: Beyond hi-tech hidebound systems. In *Knowledge Management*, Srikantaiah, T. K. and Koenig, M. E. D. (eds.), Medford, NJ: Information Today.

[11] Golfareelli, M., Rizzi, S. and Cella, L. (2004). Beyond data warehousing: What's next in business intelligence? *Proceedings of DOLAP-04*, Washington, DC, USA. www.acm.org (accessed May 17, 2006).

[12] Gangadharan, G. R. and Swamy, N. S. (2004). Business intelligence systems: Design and implementation strategies. *Proceedings of 26th International Conference on Information Technology Interfaces*, Cavtat, Croatia. http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber?=?1372391 (accessed March 15, 2007).

[13] Gangadharan, G. R. and Swamy, N. S. (2004), Business intelligence systems: Design and implementation strategies. *Proceedings of 26th International Conference on Information Technology Interfaces*, Cavtat, Croatia. http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber?=?1372391 (accessed March 15, 2007).

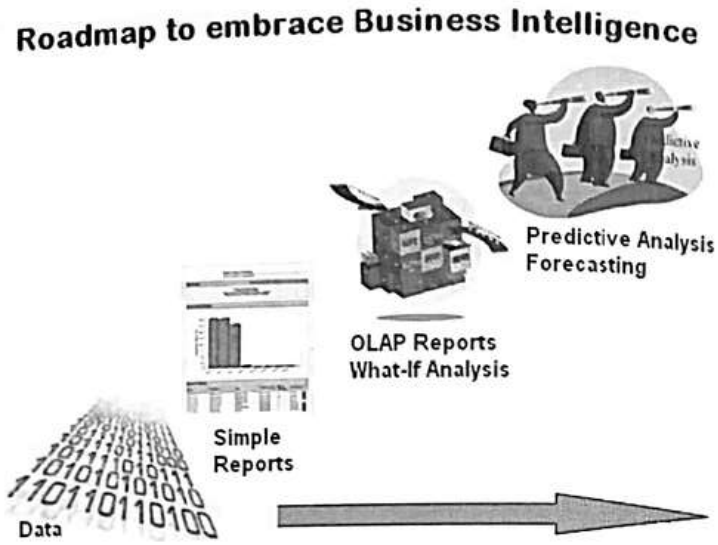


Figure 10.3: Roadmap to Business Intelligence.

Source: Analytics works (2006).[14]

Figure 10.3 sketches the evolution of business intelligence from simple data to support business decision-making.

10.1.1 What is Business Intelligence?

Business Intelligence (BI) is about getting the right information to the right decision-makers at the right time. It is an enterprise-wide platform that supports reporting, analysis and decision-making. BI leads to fact-based decision-making and a "single version of the truth." It also includes reporting and analytics. It facilitates organizational improvement through providing business insights to all employees leading to better, faster, more relevant decisions in the areas of:

- Advanced Analytics.
- Self Service Reporting.

[14] Analyticsworks is a professional services firm that helps medium and large enterprises leverage the business intelligence to make better decisions. http://www.analyticsworks.com

- End-User Analysis.
- Business Performance Management.
- Operational Applications.
- Embedded Analytics.

Business Intelligence is the process of intelligence gathering applied to business. This can be viewed as consisting of four stages in business intelligence development. The first three steps are often repeated several times before the fourth step is taken and the cycle completed, as shown in Figure 10.4.

The term "Business Intelligence" and its acronym "BI" are so pervasive in today's data-intensive lexicon that it's a challenge to know just what to make of it. If you add in all the new trendy terminology such as business process management (BPM), data mining, data warehousing, business process automation, decision support systems, query and reporting systems, enterprise performance management, executive information systems (EIS), business activity monitoring (BAM), modeling and visualization,

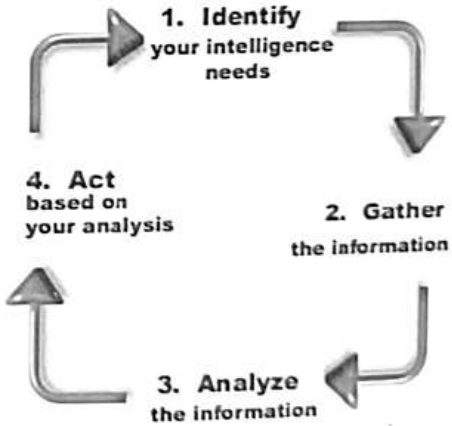


Figure 10.4: Business Intelligence Stages.

Source: FAQ (2006).[15]

[15] FAQ is to provide a one-stop Business Intelligence service business be the collection, analysis or distribution of information. http://www.faq.co.za/genl/index.shtm Copyright © 2000–2006

and so forth, your head can start spinning. A Technology Evaluation Center article by Mukhles Zaman entitled in "Business Intelligence: Its Ins and Outs" said in its Technology Evaluation 10 January 2005 issues. BI is neither a product nor a system. It is an umbrella term that combines architecture, applications, and databases. It enables the real-time, interactive access, analysis, and manipulation of information, which provides the business community with easy access to business data. BI analyzes historical data — the data businesses generate through transactions or by other kinds of business activities — and helps businesses by analyzing the past and present business situations and performances. By giving this valuable insight, BI helps decision-makers make more informed decisions and supplies end users with critical business information on their customers or partners, including information on behaviors and trends." To put it in simpler terms, BI makes the right information available in the right format to the right person at the right time. There are a variety of tools enabling BI including:

- Query and Reporting Tools —
  - Typically desktop tools that allow users to access networked databases to do basic analysis and reporting.
- Online Analytical Processing (OLAP) tools —
  - Server-based analysis tools that simulate Multi Dimensional Databases.
  - Allow users to organize data in multiple hierarchies.
- Enterprise Business Intelligence Suites —
  - Integrated Suite for Query, reporting and OLAP tools.
  - Scalable.
  - Extends to internal users, customers, suppliers and general public.

We can use BI in many areas including:

- Competition analysis.
- Customer analytics.
- Enhanced Customer Relationship Management.
- Better decision-making.
- Market examination.
- Risk scrutiny.
- Site selection.
- Territory management.

10.1.2 Business Intelligence System Architecture

A phrase coined by (or at least popularized by) Gartner Group defines BI as any computerized process used to extract and/or analyze business data. The organization engenders Business Intelligence (BI) by entering raw transactional data into a Data Mart or Data Warehouse, and generates decision support systems capabilities by means of Query, OLAP and Data Mining access tools. Components of BI can include Data Mart, Data Warehouse, OLAP and Data Mining models which are described in the following paragraphs. The architecture of BI is shown in Figure 10.5.

A successful BI architecture has four parts, as given below:

1. **Information Architecture:** The information architecture defines what business application systems are needed to access, report, and analyze

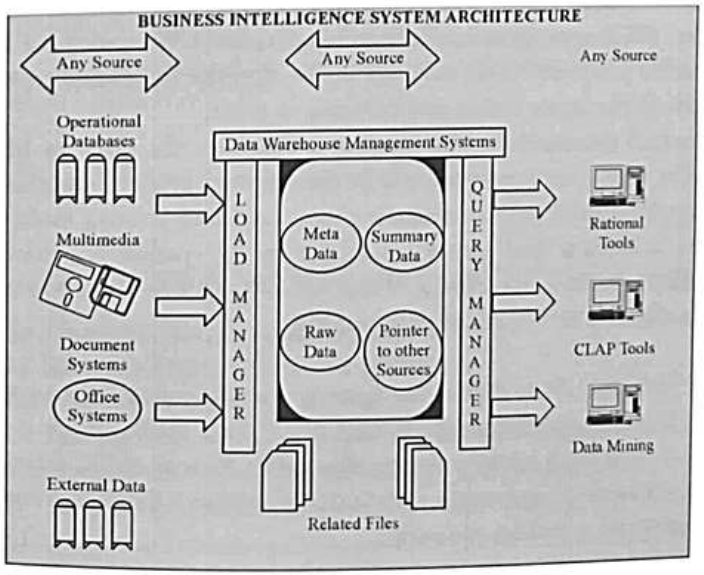


Figure 10.5: Business Intelligence Architecture.

Source: Dataquest.[16]

[16] DQIndia website is the online version of the print magazine Dataquest, referred to as the Bible of Indian IT. http://dqindia.ciol.com/.

information to enable business decision-making. It includes BI systems and business analytics applications used for reporting and analysis.

2. **Data Architecture:** The data architecture defines the data, source systems, and framework for transforming data into useful information. It begins with the sources (the information provider) and ends with the business user (the information consumer). It includes defining information requirements; determining the source of data; defining business rules and transformations; and moving, storing, formatting, and enabling access to the data. Transformations are used to create dimensions, i.e., create a common product, customer, and employee reference data; reformat data between source systems to enable common processing; handle data quality issues; and prepare data content, context, and structure for reporting and analysis.
3. **Technical Architecture:** The technical architecture defines the technology of the products and infrastructure. Technologies include RDBMS, online analytical processing (OLAP), BI, extract, transform, load (ETL), enterprise application integration (EAI), enterprise information interfaces (EII), networks, OSs, and the interfaces/protocols/APIs in the layers within and in between them.
4. **Product Architecture:** The product architecture includes the BI software, which are a combination of data-capturing tools, analysis-and-reporting tools, data-warehousing tools, and data-mining tools. Some of the BI and DM software available in the market are Intelligent Miner by IBM, Enterprise Miner by SAS, Oracle Data Mining by Oracle, and SPSS Data Mining by SPSS.

Apart from the above definition there is one other way of interpreting Business intelligence architecture. This is a theoretical BI sketch given in Figure 10.6 which is constructive for illustrating how all the BI technology fits simultaneously, and assists with both the approach definition and consequent implementation planning.

10.1.3 Business Intelligence Vendors

Many products are used in business intelligence. Some of these are listed in Table 10.1.

Other vendors include companies displayed in Figure 10.7.

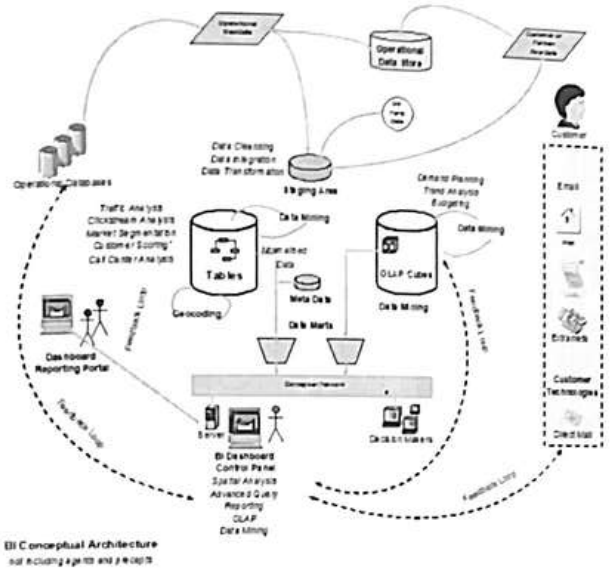


Figure 10.6: BI Conceptual Architecture.

Source: Electrosmart Ltd (2008).[17]

Some features of various vendors include:

1. SAP

SAP NetWeaver 2004s BI has the following most important new features:

- The Enterprise Services Architecture (ESA).
- The Enterprise Data Warehouse (EDW).
- Real-time data warehousing (DWH).
- Information lifecycle management and the use of nearline storage.
- Clustering, partitioning (repartitioning), and remodeling functions.
- The new extraction, transformation, and loading (ETL) process including transformation rules and data-transfer process (DTP).
- The Business Intelligence Accelerator (BIA) including its search and classification functions (TREX).

[17] The BI vision is documented in a BI strategy document to ensure that implementation of specific technology or a data structure remains focused on the BI objectives for a particular organisation. http://www.thebusinessintelligenceguide.com/bi_strategy/BI_Strategy_Document.php.

Table 10.1: Some BI Vendors.

Name	Products	Web Site
Comshare	Analytical business applications for business performance analysis and improvement.	www.comsharc.com
CorVu	Balanced scorecard and business intelligence solutions.	www.corvu.com
Cognos	Business intelligence tools and application development tools.	www.cognos.com
Hyperian	Enterprise applications for budgeting, forecasting, consolidation, and analysis.	www.hyperion.com
IQ	Enterprise reporting tools offering desktop and production reporting functionality for business and IT professionals.	www.iqsc.com
Seagate	Business Intelligence products, including Crystal and Holos.	www.seagate.com

Source: Support for the Strategists (1998).[18] *Accountancy International*, 50–53.



Figure 10.7: List of additional BI vendors.

Source: Avocado Security (2007).[19]

[18] Integrated Information System (IIS) and *Management Accounting*, www.cbs.dk

[19] Avocado Security is the first provider of cross-platform BI & Security On-Demand service. Avocado Security's business intelligence optimization platform help clients reach new untapped potential. To enable security and business intelligence information sharing across the organization, Avocado Security provides a web based On-Demand platform. http://www.avocadosecurity.com

- Advanced analytics applications.
- BI-integrated planning.
- Composite Application Framework (CAF) and barrier-free applications including.
- Visual Composer.
- Data Warehousing Workbench (DWB).
- NetWeaver architecture and within the closed-loop process (see Figure 10.8).

2. Microsoft

- Make it affordable:
  - ➢ Centers around a belief that today's BI tools are specialized and expensive.
  - ➢ Microsoft will focus on affordability and lower TCO.

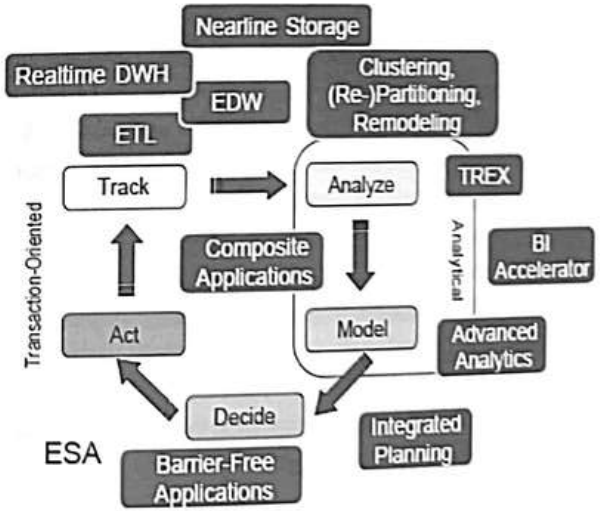


Figure 10.8: The Most Important Features of SAP NetWeaver 2004 BI.

Source: SAP NetWeaver 2004s Business Intelligence.[20]

[20] SAP NetWeaver is SAP's integrated technology platform and is the technical foundation for all SAP applications since the SAP Business Suite SAP NetWeaver is marketed as a service-oriented application and integration platform. SAP NetWeaver provides the development and runtime environment for SAP applications and can be used for custom development and integration with other applications and systems. www.sap.com/platform/netweaver/index.epx

- Leverage the Microsoft investment already made by clients:
  - ➢ End-to-end solution leveraging Microsoft SQL Server™, Microsoft Office®, Microsoft Office SharePoint® Server and Performance Management Applications.
  - ➢ Focus on the end-user's experience with Microsoft Office and the desire to "Stop Building Tools for Analysts."
- Long-term Commitment:
  - ➢ "We are dead serious about Bi," — Steve Ballmer, CEO, Microsoft Microsoft BI Conference, 11 May 2007, Seattle, WA.

Dan Vesset of IDC has commented: "Microsoft's BI Tools revenue growth in 2005 was more than 25 percent, growing at more than twice the rate of the overall market. … Microsoft's impact on the BI tools market cannot be overemphasized. This impact will mark an evolutionary change that has been put into motion by the database vendors overall and will reshape the BI tools market over the next 15 years."

Microsoft's BI tools are represented by Figure 10.9.

3. Access Group, LLC

Patrick Gaughan formed Access Group, LLC (originally named Access Services) in 1981 when computers were common only in very large companies. Because of the prohibitive cost of small business systems at that time, Access started as a service bureau and time-sharing service. They processed information for clients on customized systems, either by picking up input and delivering output, or supplied clients with access to their mainframe system via remote terminals and printers. As shared servers, personal computers and networks, and open business applications became more available and affordable, Access adapted. They grew to provide full design, supply, support, customization and service for a broad scope of business hardware and software, with a specialty niche in manufacturing systems. Now, close to 25 years later, almost every skill, tool and product used by Access Group has changed — most of them numerous times as in the following:

- Every client is a partner.
- Every need must be addressed with a combination of experience and a fresh perspective.

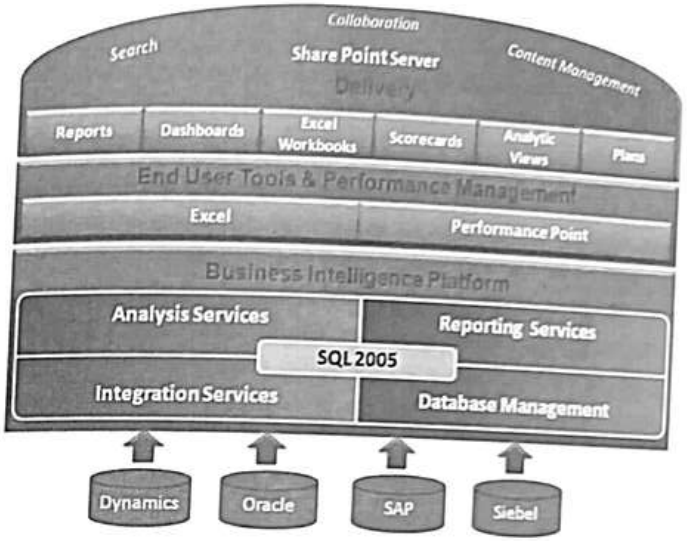


Figure 10.9: Microsoft BI Architecture.

Source: Microsoft Dynamics (2008).[21]

- Every challenge requires a quick response and thorough solution.
- Every user must be a reference.
- Its pride must result directly from what its clients achieve.

Since 1981 Access Group has assisted hundreds of organizations in installing and utilizing computer systems and technology to improve their performance and to enhance their competitive edge. Access Group has matured and prospered as its clients have succeeded. They advise, guide, train and assist clients throughout changes in the essential systems and best practices that add to their success. Access Group continues to press forward and transform its capabilities, tools, and services to support the varying needs of its clients. For a quarter of a century, in an industry too often known for missed dates and over-sold promises, it has upheld its core value — Access Builds Success.

[21] Microsoft Dynamics is a line of business software owned and developed by Microsoft, though the percentage products were originally created by other companies and known by various other names. Dynamics was earlier known by the codename "Project Green". www.microsoft.com/dynamics

4. Cognos

Cognos, an IBM company, is the world leader in business intelligence (BI) and performance management solutions. They provide world-class enterprise BI, planning and consolidation software and services to help companies plan, understand and manage financial and operational performance. Cognos brings together technology, analytical applications, best practices, and a broad network of partners to give customers a complete performance solution. They provide an open and adaptive solution that leverages an organization's ERP, packaged applications, and database investments. The system gives customers the ability to answer the questions: How are we doing? Why are we on or off track? What should we do about it? and enable them to understand and monitor current performance while planning future business strategies. Cognos serves more than 23,000 customers in more than 135 countries. Cognos was acquired by IBM in January 2008.

The important point with respect to Figure 10.10 is that Cognos 8 is the only BI product that delivers the entire range of BI capabilities

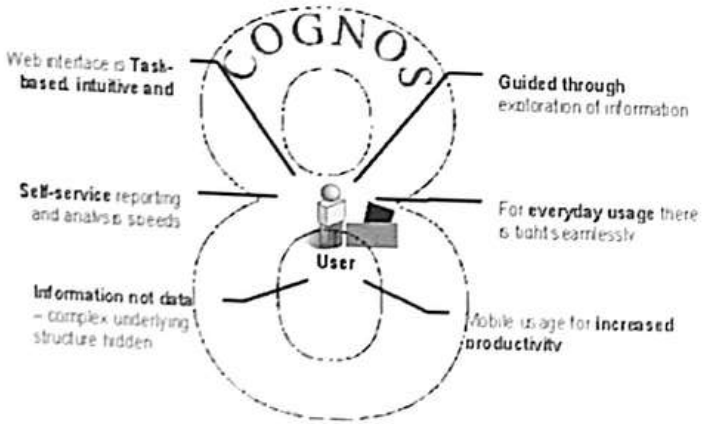


Figure 10.10: Cognos 8 Business Intelligence.
Source: LSA Solutions (2005).[22]

[22] LSA was founded in 1996 and is a services-focused, solution specialist Cognos Partner for the Cognos Performance Management suite. LSA Solutions specializes in enterprise-wide Corporate Performance Management (CPM) solutions that help businesses to align business strategies with their execution, measurement and communication. http://www.lsa-solutions.co.uk

covering: Business reporting, Business analysis, Score carding, Dashboards, Business event management etc.

5. Analyticsworks

Analyticsworks is a professional services firm that helps medium and large enterprises leverage business intelligence to make better decisions. Analyticsworks is founded by an experienced team of e-Commerce, retail, consumer finance, technology professionals with years of data analysis and strategy experience. They specialize in initiating business intelligence practice within organizations and delivering valuable solutions generating results quickly, creating immediate and lasting value. Key differentiators include in-depth understanding of vertical relationships, bringing the concept of centers of excellence within the analytics domain and using the best class open source technologies to justify ROI and minimize the total cost of ownership of the BI implementation. Analyticsworks has a high-energy environment that encourages continuous learning. Analyticsworks blends the best of a consulting firm and technical savvy of a service company with the innovative spirit of entrepreneurs.

10.2 DATA STORAGE SYSTEMS

The advent of digit technology has brought with it an increased demand for data storage. There has been tremendous progress in computer storage of data. This area of technology is clearly one where the capacity to do things has outstripped the ways in which business has been able to apply them. However, many applications are being found for this new capacity. One major user of massive storage capacity is EIS systems, which have large storage requirements due to their comprehensive nature.

Accurate information in real time facilitates expansion. Data Storage is the structure that stores private or community data for trouble-free access at whichever time and place. A data-storing and repossession system is consistent, well-organized, error-free and fast, serving diverse applications. To deal with the difficulty of supporting diverse needs,

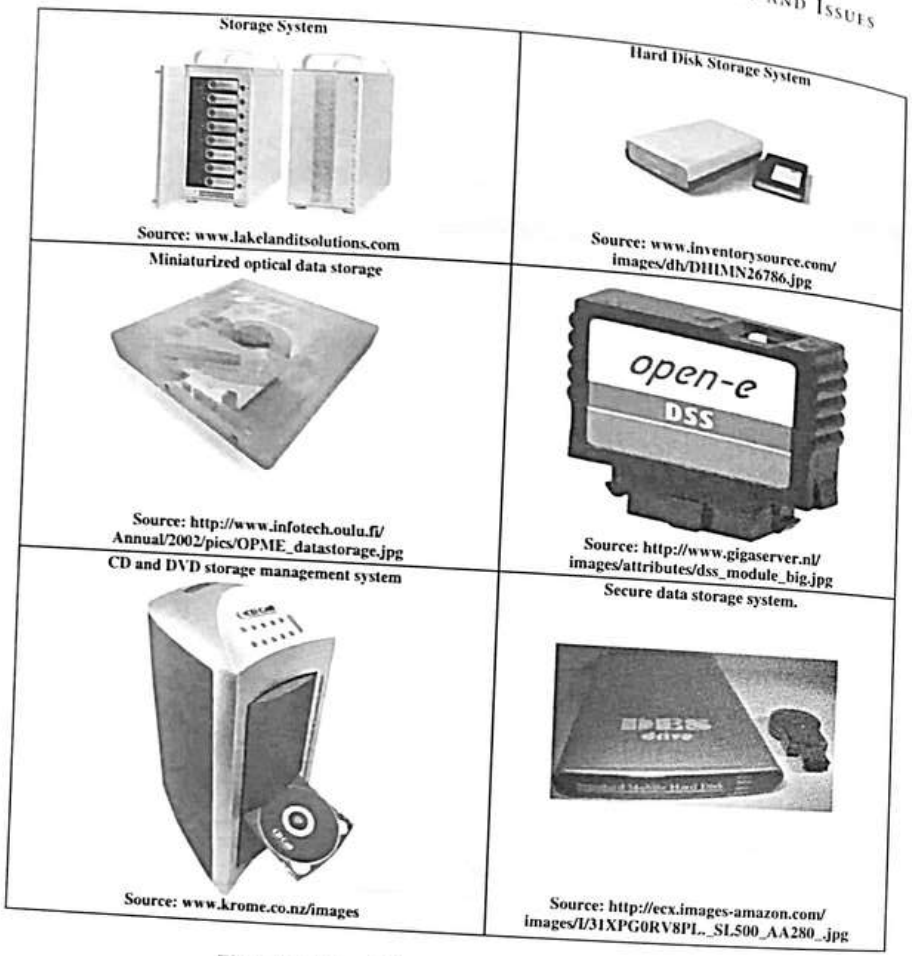


Figure 10.11: Diverse Data Storage Systems.

various categories of data storage systems are accessible including Data Backup Storage, Offsite Data Storage, Media Data Storage and many others as shown in Figure 10.11.

10.2.1 Data Warehousing

Data warehousing is an orderly and accessible repository of known facts and related data that is used as a basis for making better management decisions. Another more complete definition is: "a subject-oriented, integrated, time-variant, and non-volatile collection of data in support of

management's decision-making process."[23] Data warehouses provide ready access to information about a company's business, products, and customers. This makes it possible to organize data by subject rather than by process. These systems are used to store massive quantities of data from a variety of sources in an integrated way. The data is identified by a particular time period. Non-volatile means that the data added to the data warehouse is stable after initial formatting and cleaning, and not removed. An additional feature is efficiency, allowing quick retrieval of specific types of data. EISs generate massive amounts of data, and often data warehouses are used to support EISs. Data warehouse systems generally store data in fine granular form which other related systems such as data marts, OLAP, and other forms of decision support systems can summarize or aggregate. Data warehouses also include a metadata repository holding data about the data stored, which ensures data integrity and speedy retrieval. They also have extraction/transformation/loading (ETL) tools enabling data extraction for specific applications. Figure 10.12 shows data warehouses as a centerpoint for storage for many systems.

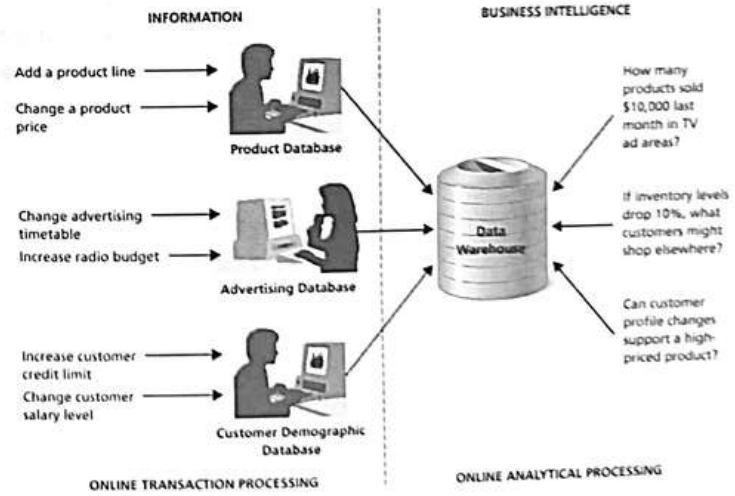


Figure 10.12: Data warehouse linkages with Information and BI.

Source: Exonous (2004).[24]

[23] Inmon, W. (2000). Data mart does not equal data warehouse. DMReview.com.
[24] http://exonous.typepad.com/mis/business_intelligence.jpg

Within data warehouses, data is classified and organized around subjects meaningful to the company such as customers, employees, or products. The data is gathered from operational systems (in addition to EIS generated data, data can be generated by bar-code readers at cash registers, information from e-commerce, daily reports, etc.) and external data sources (industry volumes, economic data, etc.). Data from different sources (shipping, marketing, billing) are integrated to a common format. Transformation also includes filtering data to eliminate unnecessary details, cleaning the data to eliminate incorrect data or duplications, and consolidating data from multiple sources. This transformation, part of the data warehouse management rather than the data mining process, makes accessing data more efficient. To demonstrate these concepts, consider toys sold over the Internet. Many toys are sold each day, with detailed information kept on each sale. The company is organized into regions, and it assigns an item number to each product. Lot numbers are assigned by the source. The data warehouse would deal with **granular** data, information in its rawest form. Within the data warehouse, each transaction may be recorded. A small portion of a specific date (Julian date 1131 — the standard industry reference to the dating system of year digit and sequential day of that year — the 131st day of year 2001) is shown in Table 10.2.

The data warehouse's purpose is permanent storage of detailed information. This repository is a reliable source of detailed information. Data entered into a data warehouse needs to be processed to ensure that it is clean, complete, and in the proper format.

10.2.2 Data Marts

Data warehouses are intended as permanent storage facilities. Data marts can exist in a number of different forms. Three of these are:[25]

1. Data marts created with a subset of data warehouse information, usually focusing on information needed by a specific set of users.

[25] Browning D. and Mundy, J. (2001). *Data warehouse design considerations.* Microsoft Corporation, www.msn.com.

Table 10.2: Data Warehouse Example.

Key	Date	Customer	Name	City	Region	Product	Item #	Quant	Lot #	Price	Source	Cost
7332	1131	C129	Kim	Seoul	Intl	Cell phone	B019	120	XY482	39.99	Samsung	10.50
7333	1131	C320	McGraw	Santa Fe	SW	IPOD cord	A001	1	BA2441	18.62	Ericsson	2.50
7334	1131	C320	McGraw	Santa Fe	SW	Camera phone	A059	1	CAB12	59.95	Motorola	13.25
7335	1131	C320	McGraw	Santa Fe	SW	Phone charger	B008	1	F431	8.62	Ericsson	0.62
7336	1131	C289	Terry	Richmond	SE	Nokia decal	C373	576	AZ26	3.59	Nokia	0.25
7337	1131	C151	Doyle	New York	NE	IPOD cord	A001	20	BA436	18.62	Ericsson	2.50
7338	1131	C151	Doyle	New York	NE	Phone charger	B008	200	F431	8.62	Ericsson	0.62
7339	1131	C238	Devlin	Austin	SW	Camera phone	A059	3	CAB12	59.95	Motorola	13.25
7340	1131	C241	Kelly	Orlando	SE	Camera	A007	1	RA69	231.12	Motorola	36.67

2. Free-standing data marts independent of data warehouses, making them a quick and less expensive (although less powerful) means of implementing the data warehouse idea.
3. A prototype for a future full-scale data warehouse.

Data, once stored in a data warehouse, is usually not changed without a compelling reason. In order to apply data mining, an intermediate storage form is used. Data marts are sometimes used to extract specific items of information for data mining analysis (type 1 above). Data marts have a number of advantages. First, they are available for data miners to work with, transforming information to create new variables (such as ratios, or coded data suitable for a specific application) without fear that these transformations will contaminate the data warehouse. Second, only that information expected to be pertinent to the specific data mining analysis is extracted. This vastly reduces the computer time required to process the data, as data marts are expected to contain small subsets of the data warehouse's contents and to have ample space available to generate additional data by transformation. Possible data warehouse linkages are displayed in Figure 10.13.

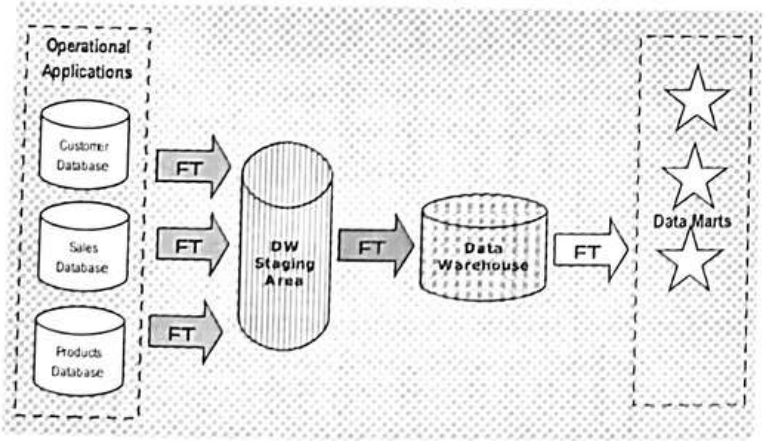


Figure 10.13: Data warehouse linkages with Information and BI.
Source: Data-warehouses.net[26]

[26] data-warehouses.net/architecture/datamarts.html

A data mart is interested in data relating to a question under study. For instance, a firm might be interested in examining the characteristics of customers who buy their products. The firm would be interested in response to advertising by mail, advertising by Internet, or advertising by print media. The firm would be interested in who buys by region, as well as the profitability by product, and the effectiveness of advertising in increasing sales by product. The data warehouse would contain much useful information related to these questions.

As with all centralized data systems, technical expertise is required to operate a data mart. However, there is a slight difference in its application from that of a data warehouse. A data warehouse would have permanent staff to operate that system, with an ongoing responsibility to clean and maintain data, and to ensure data integrity and security. A data mart, on the other hand, is typically applied to shorter-term projects, usually with the intent of collecting data for a specific study.

Analysts have choices to make about manipulating currently available information, generating additional information, or purchasing information from vendors. Information available from the data warehouse could include sales volume and profitability figures by time and region. Data marts could extract this data, aggregate it in a form useful for data mining, and keep only that information important for the study at hand (and possibly specify a time frame). But the data warehouse does not include the advertising media that triggered sales. This information would have to be generated by survey. Survey results could be added to the data mart. For observations where survey information is not obtained, the observation might be deleted, or it might be retained with a code for missing data. Additional observations or variable information might be available by purchase from information vendors. A subset of data mart entries for non-vehicular toy products is shown in Table 10.3.

Using raw data from the data warehouse, profit rates are calculated based on multiplying volume by the difference between price and cost, and inserting this in the appropriate data mart column. Customers listed in Figure 10.2 were contacted by survey to identify their media of product introduction. Survey results are appended within the data mart. The data warehouse provided the first five observations presented. External sources were used to supplement the data warehouse results. The last four

Table 10.3: Example of a Data Mart.

Customer #	Region	Item #	Qty.	Profit	Source	Media
C320	Southwest	A001	1	16.12	Ericsson	Internet
C320	Southwest	B008	1	8.00	Ericsson	Print
C289	Mideast	C373	576	3075.84	Nokia	Internet
C151	Northeast	A001	20	322.40	Ericsson	Internet
C151	Northeast	B008	200	1600.00	Ericsson	Internet
X	Northeast	D412	24	1248.00	Vendor	Internet
X	Midwest	B429	1	8.77	Vendor	Mail
X	Northwest	B231	12	36.36	Vendor	Print
X	Northwest	A622	1	17.25	Vendor	Print

observations presented are from this source, and they have no customer number. The items from these sources are keyed to this organization's catalog of products, or to the nearest equivalent. Profit is estimated for these additional entries based on this organization's prices and costs. The purpose of the data mart here is to feed data mining.

10.2.3 On-Line Analytic Processing

Another type of application to exploit data warehouses might not be as primitive in terms of the analysis conducted, but is tremendously fast in response and enables executives to make timely decisions: on-line analytical processing (OLAP). OLAP systems are multi-dimensional databases. These systems allow analysts to display data in one or more of a number of different dimensions, such as time, geographic region, product, organizational department, customer, or other factors. While data warehouses focus on efficiently storing vast quantities of data, OLAP systems are designed to make it easy to analyze data.

OLAP systems allow firms to deliver access to data and report generating tools throughout their organization in virtual time. The ability to access OLAP packages has allowed Lockheed Martin Tactical Aircraft Systems to make data widely available for analysts. The OLAP system provided a single source to find program and business-management metrics, staffing information, risk and technical measurements, sales and cost forecasting, and overhead information.

An OLAP application would focus more on analyzing trends or other aspects of organizational operations. It may obtain much of its information from the data warehouse. The OLAP application extracts granular information that is of interest to the users being supported, aggregates this information, and makes the information easily accessible on a number of dimensions. This information could be accessed to make a report by product, as shown in Table 10.4.

Tables, even if joining data from several sources, limit the review of information. Often, executives need to view information in multiple combinations of two dimensions. For example, an executive might want to see a summary of the quantity of each product sold in each region. Then, she might want to view the total quantities of each product sold within each city of a region. In addition, she might want to view quantities sold of a specific product in all cities of all regions. OLAP is specially designed to answer queries such as these. OLAP applications let a user rotate virtual "cubes" of information, whereby each side of the cube provides another two dimensions of relevant information. Figure 10.14 displays the operation of an OLAP system.

The purpose of OLAP here is summarizing data with a report focus. OLAP products have spreadsheet computational capability, as well as organization of data by selected dimensions (by region, by quarter).

10.2.4 Data Quality

Data warehouse projects can fail for a number of reasons. One of the most common reasons is users' refusal to accept the validity of data obtained from a data warehouse (Wu, 2000).[27] This is an issue of data integrity. Such problems can arise because of one or more of the following:

- Corruption of data or missing data from the original sources.
- Failure of the software transferring data into or out of the data warehouse.
- Failure of the data-cleansing process to resolve data inconsistencies.

[27] Wu, J. (2000). Ensuring data integrity (Parts 1, 2, and 3). *DM Review*. www.datawarehouse.com.

Table 10.4: Example of an OLAP.

Product	Region	Q IV Last Year	Quarter I	Quarter II	Quarter III	Quarter IV	Revenue This Year	Profit This Year
Cell phone	Northeast	326	122	98	82	125	15,235	6,875
	Southeast	258	68	42	39	51	7,536	3,168
	Midwest	412	151	112	106	165	19,871	8,168
	Southwest	168	39	26	25	32	4,536	1,832
	Northwest	151	36	31	30	35	4,986	2,126
	International	56	65	73	86	92	11,965	3,965
	Total	1,371	481	382	368	500	64,129	26,134

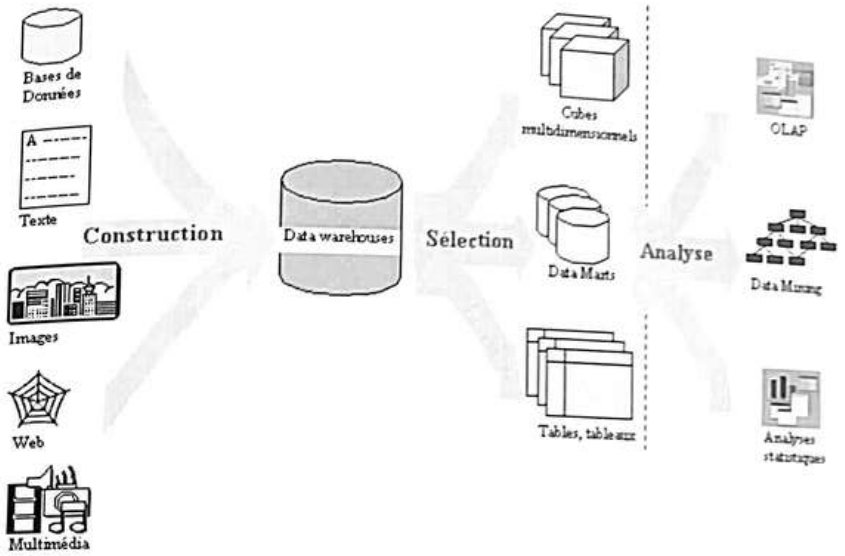


Figure 10.14: Execution of OLAP.

Source: www.bdd.univ-lyon2.fr

Figure 10.15 shows the integration of quality in the data gathering process.

In the initial stages of data warehouse use, it is critical that data proves to be reliable. Once a reputation is lost, it is very difficult to recover. The information system staff operating the data warehouse must verify the integrity of data, ensuring that when data is loaded into the data warehouse, it is stored as planned. It is also necessary that the systems used to extract data from the data warehouse function properly. This is accomplished through cleaning up new data to do things such as removing redundancies, filling in blanks and missing fields, and organizing data into consistent formats.

10.2.5 Data Integrity

Data integrity requires that meaningless, corrupt, or redundant data not be entered into the data warehouse. Controls can be implemented prior to

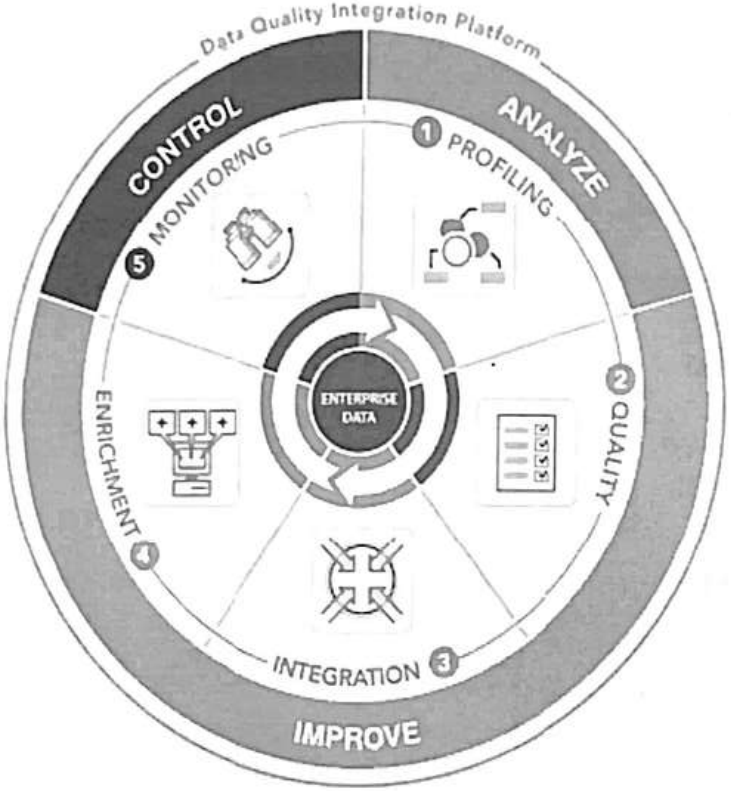


Figure 10.15: Data Quality Integration Platform.

Source: 1105 Media, Inc. (2005).[28]

loading data, in the data migration, cleansing, transforming, and loading processes. This is the most efficient stage to prevent meaningless, corrupt, or redundant data from entering the system. Figure 10.16 shows a schematic representation of IT control aimed at attaining data quality.

An example of multiple variations for two of the variables in the example of a data warehouse is shown in Table 10.5.

[28] TDWI (The Data Warehousing Institute™) provides education, training, certification, news, and research for executives and information technology (IT) professionals worldwide. Founded in 1995, TDWI is the premier educational institute for business intelligence and data warehousing.

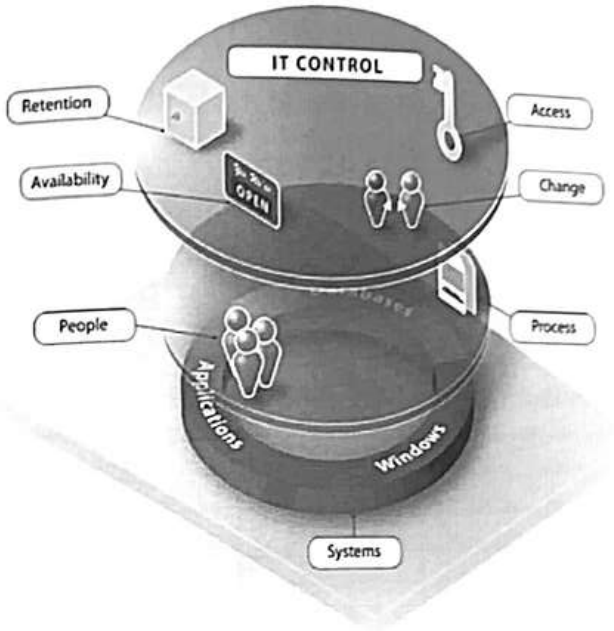


Figure 10.16: Data Integrity through Quality Control.

Source: www.kccaretech.com.[29]

In this case, the name variable includes three variations of the same customer. The second is a misspelling. The third is correctly spelled, but it includes a more complete definition. Since the name "Kim" is fairly common, it would probably be best to use the form "JD Kim," although this decision would be dependent upon the use of the variable. The process of developing unique variable values is data standardization.

10.2.6 Matching

Matching involves associating variables. For instance, in this database, the variable "Customer" is clearly associated with a unique "Name." We can see that the Customer variable includes a misread value (C729 rather than

[29] Kccaretech provides full support in billing and claims services. www.kccaretech.com

Table 10.5: Extract from Database for Name Kim.

Key	Date	Customer	Name	City	Region	Product	Item #	Qty.	Lot #	Price	Source	Cost
7332	1131	C129	Kim	Seoul	Intl	Cell phone	B019	120	XY482	39.99	Samsung	10.50
7671	1131	C129	Kimm	Inchon	Intl	Cell phone	B019	12	ZZ1243	41.16	Ericsson	10.66
7822	1131	C729	JD Kim	Seoul	Intl	Cell phone	B019	144	TU642	37.65	Samsung	10.12
7865	1131	C129	Kim	Seoul	Intl	Cell phone	B019	1	VV336	38.86	Samsung	8.67

C129) in the third row. It is often difficult to detect such errors, especially in conjunction with other problems such as variations in name spellings.

It is necessary for efficient data warehouse operation that the database contain the minimal number of consistent entries for each variable. As discussed above, one name entry for "JD Kim" needs to be identified. The overall system needs to be adjusted to reflect these choices. Software used to introduce new data into the data warehouse needs to check that the appropriate spellings and entry values are used. This also includes matching companies with addresses, an obvious opportunity for variety. Care needs to be taken to keep up with changes, such as telephone area code changes, or new zip codes. Personnel turnover makes maintaining proper contact names an interesting challenge. Means to guarantee data quality begin with ensuring that the data extraction process operates correctly. A framework for error identification and correction as well as reconciliation needs to be operating when the data warehouse is created. Data validation and testing tools are needed to monitor data quality and resolve problems as they arise.

Once data is stored in the data warehouse, controls can be applied to detect accuracy and completeness. Quick reviews should be performed soon after data is loaded to make sure that the correct numbers of records were loaded. It is useful to check aggregate totals as a means of verifying a degree of accuracy. More detailed validation efforts are often performed during data warehouse implementation. Ownership and accountability for particular data are assigned to a specific person or organization. Detailed validation checks whether data is complete and correct, whether business rules are followed, and whether the transformation processes of consolidation, filtering, cleaning, and aggregating are done properly. Validation also checks to make sure that data was loaded correctly. Data also is checked to make sure that entries are within tolerance levels. Any errors detected should be investigated to determine cause, so that appropriate changes can be made to the overall system. Data marts are products designed to select particular data from data warehouses (as well as from external sources) to be used for analysis, especially data mining. OLAP products come in a variety of product forms, but all are intended to give users the ability to design reports that give them insight into their operational environment. Table 10.6 compares the three database products discussed in broad terms.

Table 10.6: Comparison of Database Products Related to EIS.

Product	Use	Duration	Granularity
Data warehouse	Repository	Permanent	Finest
Data mart	Specific study	Temporary	Aggregate
OLAP	Report and analysis	Repetitive	Summary

Data quality is very important in ensuring the accuracy needed for successful system use. Data needs to be checked for accuracy before entry into the data warehouse. In an ideal system, if accurate data is entered, there should be little problem during subsequent operations. Realistically, there are many required changes to data, which makes administration of a data warehouse challenging. Data warehouses are capable of storing vast quantities of data. However, their implementation is not trivial. Missing and miscoded data have to be cleaned up, and variables often come in a variety of types, such as nominal data with no numeric content, dates, counts, averages, and many other forms. Relationships may be difficult to identify because of weak relations, often masked by other relationships. There are a number of publications available to explain more about data warehouses.[30] Now that we have established that the vast amounts of information generated by EIS systems can be efficiently stored and retrieved, we turn to ideas about how to utilize this information.

10.3 DATA MINING OVERVIEW

Data mining refers to the analysis of the large quantities of data that are stored in computers.[31] For example, grocery stores have large amounts of data generated by purchases. Bar coding has made grocery checkout very convenient. Grocery stores and other retail stores are able to quickly

[30] Inmon, W. H. (2005). *Building the Data Warehouse*, 4th ed. New York: Wiley; Silvers, F. (2008). *Building and Maintaining a Data Warehouse*. Philadelphia: Auerbach.

[31] Berry, M. J. A. (2002). *Mastering Data Mining*. New York: John Wiley & Sons; Olson D. L. and Shi, Y. (2007). *Introduction to Business Data Mining*. Boston: McGraw-Hill/Irwin.

process purchases, and use computers to accurately determine product prices. These same computers can help the stores with their inventory management, by instantaneously determining the quantity of items of each product on hand. They are also able to apply computer technology to contact their vendors so that they do not run out of the things that are in demand. Computers allow the store's accounting system to more accurately measure costs, and determine the profit that store stockholders are concerned about. All information is available based upon the bar coding information attached to each product. The benefits of bar coding are not only providing faster checkout service. The entire business management process can use the information generated.

Data mining can use bar code information to make better business decisions. Data mining is not limited to retail inventory control through bar coding. It is also widely used by banking firms in soliciting credit card customers, by insurance and telecommunication companies in detecting fraud, by manufacturing firms in quality control, and many other applications. One of the most prominent applications of data mining is support of **customer relationship management** (CRM). Great Atlantic & Pacific grocery stores use data mining to target customers and centralize buying. Fingerhut has become very successful in **micromarketing**, targeting small groups of highly responsive customers. Media companies such as R. R. Donnelly & Sons provide consumer and life style data, as well as customized individual publications to firms that use data mining for catalog marketing. Data mining requires identification of a problem, along with collection of data that can lead to better understanding, and computer models to provide statistical or other means of analysis. There are two general types of data mining studies. **Hypothesis testing** involves expressing a theory about the relationship between actions and outcomes. In a simple form, there is a hypothesis that advertising will yield greater profit. This relationship has long been studied by retailing firms in the context of their specific operations. Data mining can be applied to identify relationships based on large quantities of data, which could include testing the response rates to various types of advertising on the sales and profitability of specific product lines. The second form of data mining study is **knowledge discovery**. In this form of analysis, a preconceived notion may not be present, but rather be seen by looking at the data. This may be supported by

visualization tools which display data, or through fundamental statistical analysis such as correlation analysis.

Data mining has been called exploratory data analysis. Masses of data (generated from cash registers, from scanning, from topic-specific databases throughout the company) are explored, analyzed, reduced, and reused. Searches are performed across different models proposed for predicting sales, marketing response, and profit. Classical statistical approaches are modified in data mining, due to the lack of time for systematic exploration through classical statistical methods. However, some of the tools developed by the field of statistical analysis are harnessed through automatic control (with some key human guidance) in dealing with data. Data mining can be conducted with a variety of statistical and data manipulation tools. For large projects, data mining tools need to be versatile, scalable, capable of accurately predicting responses between actions and results, and capable of automatic implementation. Versatile refers to the ability of the tool to apply a wide variety of models. Scalable implies the ability to efficiently analyze large and even very large data sets. Automation is sought, but its application is relative. Some analytic functions are often automated, but human setup prior to implementing procedures is required. In fact, analyst judgment is critical to successful implementation of data mining. Proper selection of data to include in searches is critical. Too many variables produce too much output, while too few can overlook key relationships in the data.

10.3.1 Typical Benefits of Data Mining

Data mining is expanding rapidly with many benefits to business. Most of these gains involve relatively minor advantages. Two of the most profitable application areas have been by direct marketing organizations (to identify those with marginally greater probabilities of responding to different forms of marketing media) and banks (who use data mining to more accurately predict the likelihood of people to respond to offers of different services offered by the bank). Many companies are using this technology to identify their blue-chip customers so that they can provide them the service needed to retain them (customer relationship management). Figure 10.17 gives one view of the process of data mining with respect to data access.

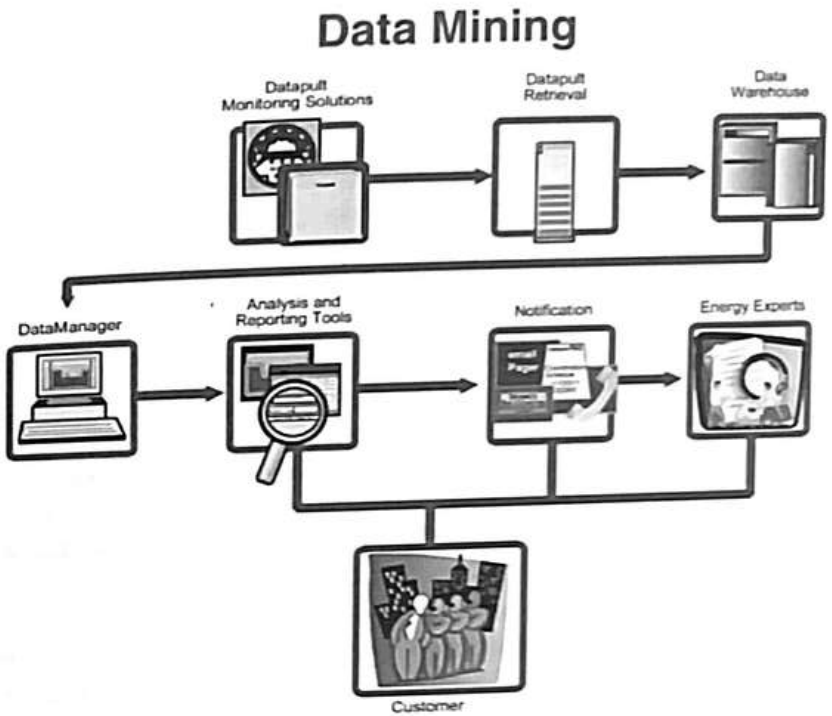


Figure 10.17: Data Mining Process linked to Data Warehouse.
Source: Datapult Limited Partnership (2000).[32]

Bank of America in San Francisco found that a small proportion of its customers determined bank profitability (20 percent in their case). Bank of America developed profiles of its top accounts to target services. They also were able to assess the likelihood that particular customers would take their business to a competitor (**churn**, in telephone business terminology, which we will see again later in this chapter).

[32] Datapult Data Mining Services transform data into information. Information that provides knowledge for reducing operating costs and minimizing equipment maintenance costs. Data Mining Services provide customers with tools to quickly sift through the clutter of vast amounts of data to identify potential areas of improvement. In addition, Datapult's capabilities include a dedicated staff of Energy Experts to help identify opportunities and to provide energy use and maintenance consultation. At the customer's option, Datapult Energy Experts will "mine" their usage data for valuable information to uncover savings opportunities.

The casino business has also adopted data warehousing and data mining. Historically, casinos have wanted to know everything about their customers. Harrah's Entertainment Inc. is one of many casino organizations that use incentive programs (Haugsted, 2007).[33] Harrah's uses data mining to weed out thrifty visitors and cultivates those who spend thousands on food, shopping and gambling. Harrah's Hotel Revenue Management System calculates how much each customer is worth to Harrah's and prices their hotel rooms accordingly. Harrah's also offers cash-back coupons as a reward for fast gambling play. About 8 million customers hold Total Gold cards, which are used whenever the customer plays at the casino, or eats, or stays, or spends money in other ways. Points accumulated can be used for complimentary meals and lodging. More points are awarded for activities that provide more profit to Harrah's. The information obtained is sent to the firm's corporate database, where it is retained for several years. Trump's Taj Card is used in a similar fashion. Recently, high competition has led to the use of data mining. Instead of advertising the loosest slots in town, Bellagio and Mandalay Bay have developed the strategy of promoting luxury visits. Data mining is used to identify high rollers, so that they can be cultivated. Data warehouses enable casinos to estimate the lifetime value of players. Incentive travel programs, in-house promotions, corporate business, and customer follow-up are tools used to maintain the most profitable customers. Casino gaming is one of the richest data sets available. Very specific individual profiles can be developed. Some customers are identified as those who should be encouraged to play longer. Other customers are identified as those who are discouraged from playing.

10.3.2 Business Data Mining Applications

There are many uses for data mining in business. A variety of techniques are applied to a diverse set of problems. See Table 10.7.

Data mining offers retailers in general, and grocery stores specifically, valuable predictive information from mountains of data. Grocery stores in

[33] Haugsted, L. (2007). Freebies help Harrah's ward off flashy rivals. *Multichannel News*, 28(30), 20; _____ (2004). Off-the-charts results. *Baseline* 31, 26–32.

Table 10.7: Data Mining Applications.

Area	Applications
Retailing	Market basket analysis, affinity positioning, cross-selling
Banking	Customer Relationship Management
Credit Card Management	Lift, churn
Insurance	Fraud detection
Telecommunications	Churn (customer turnover)
Telemarketing	On-line caller information
Human Resource Management	Churn (employee turnover)

particular generate mountains of cash register data that require automated tools for analysis. Software is marketed to service a spectrum of users. In the past, it was assumed that cash register data was so massive that it couldn't be quickly analyzed. However, current technology enables grocers to look at customers who have defected from a store, their purchase history, and characteristics of other potential defectors. Targeted marketing programs are beginning to be successfully used by grocers. Single-store operations may be able to operate with PC software for as little as $4,000. Free Internet software is emerging as well. Most larger chain operations will have to spend up to $750,000 for data mining operations.

10.4 DIFFERENCE BETWEEN DATA WAREHOUSE AND BUSINESS INTELLIGENCE

A data warehouse incorporates enterprise-wide corporate data in a single repository, whereas Business Intelligence incorporates analysis and reporting tools that utilize information available in the Data Warehouse. Business Intelligence involves the process of analyzing large amounts of data in order to extract new kinds of useful information such as implicit relationships between different pieces of data. The differentiation between Data Warehouse and Business Intelligence can be very well explained with the help of Figure 10.18.

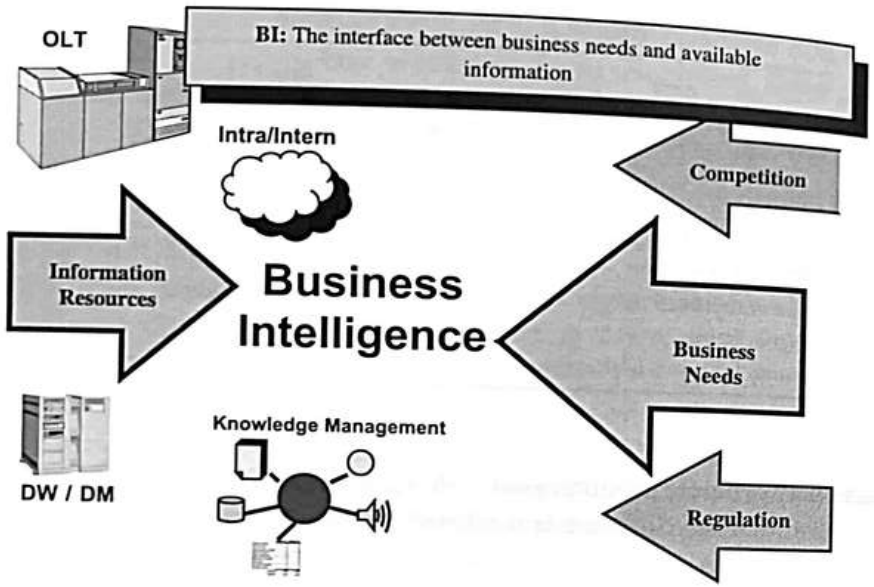


Figure 10.18: Relationship of Data Warehouses and Business Intelligence.
Source: Enterprise Group Ltd (1999).[34]

10.5 CUSTOMER RELATIONSHIP MANAGEMENT

The banking industry was one of the first users of data mining. Understanding the value a customer provides the firm makes it possible to rationally evaluate if extra expenditure is appropriate in order to keep the customer. An analyst at The Tower Group predicted that spending by U.S. banks on customer relationship management (CRM) will climb to $9 billion by the end of 1999, and grow at 11 percent per year thereafter. Deloitte Consulting found that only 31 percent of senior bank executives were confident that their current distribution mix anticipated customer needs. Figure 10.19 describes how business intelligence links organizational systems with profit-generating uses.

[34] Enterprise Group Limited (EGL) operated as a Business Intelligence (BI), technology and management consulting form from 1992 to 2003. EGL served clients worldwide, with a focus on the United States of America and Western Europe. EGL's BI practice focused on Global 1000 firms, while the technology and management practices primarily served high technology start-up companies. www.egltd.com.

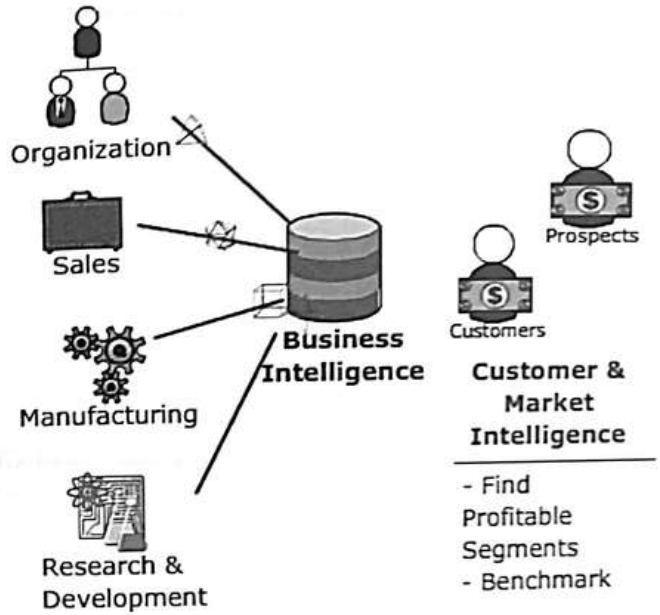


Figure 10.19: Business Intelligence as a Link between Organizations and Profitability.
Source: David Harper.[35]

Support of customer relationship management is the form of data mining most commonly associated with EIS. CRM allows businesses to identify the profitability of specific customers, and to increase chances of retaining them. This is accomplished by having all relevant information readily available that is needed for planning, product, and service throughout the customer life cycle. Oracle purchased Siebel Systems and has made that leading CRM add-on package a module in the Oracle enterprise system. SAP also supports CRM as a module in their newer versions of their enterprise system software. However, as with any innovative application of technology, there are growing pains. Ernst & Young studies noted CRM

[35] David Harper is a Contributing Editor and an Investopedia Advisor. In addition to being a writer for Investopedia, David Harper, CFA, FRM is a contributing editor for the Investopedia Advisor. David is Principal of Investor Alternatives LLC, a firm dedicated to research, consulting (e.g., derivatives valuation) and financial education. He publishes *The Bionic Turtle*, a set of study aids designed to help people prepare for financial exams.

failure rates of 70 to 80 percent, and large gaps between planned and actual benefits from CRM systems (Holland and Skarke, 2001).[36] These numbers are typical in IS/IT projects, however. As with any technology, the risks must be realized along with the opportunities to much more effectively manage organizational resources.

Fleet Financial Group has blended product and customer-based approaches. Information was being used to provide customer focus within a product-based organization rather than reorganizing around customer groups, as other financial institutions have done. Fleet has invested about $30 million in a data warehouse to support the entire organization. They also hired about 60 database marketers and statistical/quantitative analysts as well as specialists in decision support and other areas. First Union (now Wachovia) has concentrated on the contact-point end of CRM. The bank previously had very focused product groups with little coordination. First Union has created marketing customer information files, which integrated information across products through an enterprise-wide data warehouse and marketing-based data mart. Their CRM structure uses statistical tools to develop offers for customers. Data mining provides a way for banks to identify patterns. This is valuable in assessing loan applications, as well as in target marketing. Credit unions use data mining to track member profitability, as well as monitoring the effectiveness of marketing programs and sales representatives. They are also used in member care, seeking to identify what credit union customers want in the way of services.

10.5.1 Credit Card Management

The credit card industry has proven very profitable. It has attracted many card issuers, and many customers carry four or five cards. A common practice is balance surfing, where the card user pays an old balance with a new card. These are not considered attractive customers, and one of the uses of data warehousing and data mining is to identify balance surfers. The profitability of the industry has also attracted those who wish to push

[36] Holland, W. and Skarke, G. (2001). Is your IT system VESTed? *Strategic Finance, 83(6), 34–37.*

the edge of credit risk from both the customer and the card issuer perspective.

Data mining tools used by banks include credit scoring. A key is a consolidated data warehouse, covering all products, including demand deposits, savings, loans, credit cards, insurance, annuities, retirement programs, securities underwriting, and every other product banks provide. Credit scoring used to be conducted by bank loan officers who considered a few tested variables, such as employment, income, age, assets, debt, and loan history. Credit scoring applies statistical and mathematical models including many more variables on a much larger scale. Credit scoring provides a number for each applicant by multiplying a set of weighted numbers determined by the data mining analysis times ratings for that applicant. These credit scores can be used to make accept/reject recommendations, as well as to establish the size of a credit line.

Customers who usually use ATM machines can be presented with electronic sales pitches of products this particular customer is likely to be interested in. If a database indicates a new address for a customer with high credit scores, this customer may have traded up to a new, larger house, and may be a prime target for an increased credit line, a higher-end credit card, or a home improvement loan, which can be offered in a card statement mailing. Databases can also be used to support telephone representatives when customers call. The representative's computer screen can indicate the customer's characteristics as well as products the customer may be interested in.

The new wave of technology is broadening the application of database use and targeted marketing strategies. In the early 1990s, nearly all credit card issuers were using mass-marketing to expand their cardholder bases. However, with so many cards available, broad-based marketing campaigns have not been as effective as they initially were. Card issuers are more carefully examining the expected net present value of each customer. Data warehouses provide the information giving issuers the ability to try to more accurately predict what the customer is interested in, as well as their potential value to the issuer. Desktop campaign management software is used by the more advanced credit card issuers, utilizing data mining tools such as neural networks to recognize customer behavior patterns to predict their future relationship with the bank.

10.5.2 Insurance

The insurance industry utilizes data mining for marketing, just as retailing and banking organizations do. But they also have specialty applications. Farmers Insurance Group has developed a system for underwriting that generates millions of dollars in higher revenues and lower claims. The system allows the firm to better understand narrow market niches, and to predict losses for specific lines of insurance. One discovery was that it could lower its rates on sports cars, which increased their market share for this product line significantly. Farmers uses seven databases and 35 million records.

Unfortunately, our complex society leads to some inappropriate business actions, including insurance fraud. Specialists in this underground industry often use multiple personas to bilk insurance companies, especially in the automobile insurance environment. There are a number of specialty data mining software tools for various applications of fraud detection. By linking names, telephone numbers, streets, birthdays, and other information with slight variations, patterns can be identified indicating fraud. The similarity search engine has been found to be able to identify up to seven times more fraud than exact-match systems.

10.5.3 Telecommunications

Deregulation of the telephone industry has led to widespread competition. Telephone service carriers fight hard for customers. The problem is that once a customer is obtained, it is attacked by competitors, and retention of customers is very difficult. The phenomenon of a customer switching carriers is referred to as **churn**, a fundamental concept in telemarketing as well as in other fields.

A director of product marketing for a communications company considered that one-third of churn is due to poor call quality, and up to one-half is due to poor equipment. That firm has a wireless telephone performance monitor tracking telephones with poor performances (Reeves, 1998).[37] This system reduced churn by an estimated 61 percent,

[37] Reeves, B. (1998). All in the family. *Wireless Review*, 15(7), 42–50.

amounting to about three percent of the firm's overall subscribers over the course of a year. Given an average business volume of $150 per month, this churn reduction was estimated to be worth $580,000 in revenue a year. The firm's cellular fraud prevention system monitors traffic to spot problems with faulty telephones. When a telephone begins to go bad, telemarketing personnel are alerted to contact the customer and suggest bringing the equipment in for service.

Another way to reduce churn is to protect customers from subscription and cloning fraud. A number of cloning fraud prevention systems are marketed. These systems provide verification that is transparent to legitimate subscribers. Subscription fraud has been estimated to have an economic impact of $1.1 billion. Deadbeat accounts and service shutoffs are used to screen potentially fraudulent applicants. Churn is a concept that is used by many retail marketing operations. Banks widely use churn information to drive their promotions. Once data mining identifies customers by characteristic, direct mailing and telemarketing are used to present the bank's promotional program. The mortgage market saw massive refinancing in the early 1990s. Banks were quick to recognize that they needed to keep their mortgage customers happy if they wanted to retain their business. This has led to banks contacting current customers if those customers hold a mortgage at a rate significantly above the market rate. While they may cut their own lucrative financial packages, banks realize that if they don't offer a better service to borrowers, a competitor will.

10.5.4 Telemarketing

Telephone providers obviously are among the many marketing operations utilizing telemarketing. MCI Communications has utilized a strategy of **data marts**, extracting data on prospective customers from a data warehouse. This data is typically applied in a two-month program, after which the data mart is shut down. The data system required a multimillion-dollar investment in data marts and parallel hardware to support it. This operation was staffed by 45 people.

Segmentation involves grouping data with common characteristics, such as the set of customers who respond to new promotions, the set of customers who respond to discounts, or the set of customers who respond

to new product offers. This information is used to determine the group of customers offered a new service, or to predict the set of customers most likely to commit fraud. Data mining can be used to determine segments. Once segments have been defined, on-line analytic processing (OLAP) tools can be used to explore them in greater depth. The MCI system has been used for trend-spotting. If a prospect turns down one frequent-flier pitch enough times, the program can suggest a different approach, such as switching airlines. On the data mart, data can be updated by the user. Data marts for a specific application are designed to include only the information needed for the specific promotion. However, it is important that all required information be included, so initial design of the data mart is crucial. Too much data slows the system, while too little causes problems requiring additional data.

10.5.5 Human Resource Management

Business intelligence is a way to truly understand markets, competitors, and processes. Software technology such as data warehouses, data marts, and data mining or on-line analytical processing (OLAP) makes it possible to sift through data in order to spot trends and patterns that can be used by the firm to improve profitability. In the human resources field, this analysis can lead to identification of individuals who are liable to leave the company unless additional compensation or benefits are provided.

Data mining can be used to expand upon things that are already known. A firm might know that 20 percent of its employees use 80 percent of services offered, but may not know which particular individuals are in that 20 percent. Business intelligence provides a means of identifying segments so that programs can be devised to cut costs and increase productivity.

10.5.6 Data Mining Tools

There are many statistical and analytic software tools marketed to provide data mining. One of the most widely used products is SAS Institute's Enterprise Miner. But many other products also are being used. These products use one or more of a number of analytic approaches. The major

categories of methods applied are regression, decision trees, neural networks, cluster detection, and market basket analysis. The market supplying software for both data warehousing and data mining is growing very rapidly. Vendors selling data access tools include IBM, SAS Institute Inc., Microsoft, Brio Technology Inc., Oracle, and others. IBM's Intelligent Mining Toolkit has a set of algorithms available for data mining to identify hidden relationships, trends, and patterns. SAS's System for Information Delivery integrates executive information systems, statistical tools for data analysis, and neural network tools.

10.5.7 Real Practice

Our example of real practice reviews Wal-Mart's highly successful data warehouse system, which has been a key element in Wal-Mart's dominance of their business field. Wal-Mart was founded in 1962. Forty years later it dominated the retail market in the U.S. One of the primary reasons for this dominance has been the use of information technology, used in great part to support Wal-Mart's core competency of supply chain distribution over its many outlets. Wal-Mart uses a data warehouse consisting of 101 terabytes in 2001, believed to be the world's largest database.[38] The original investment for this data warehouse was over $4 billion.

The initial Wal-Mart data warehouse was stocked with point of sale and shipment data. This has been supplemented with inventory, forecast, demographic, markdown, return, and market basket information. Data about competition was also included, with over 65 million transactions processed per week. This data warehouse included 65 weeks of data by item, by store, and by day.

The purpose of this information was to support decision-making. Buyers, merchandisers, logistics personnel, and forecasters had direct access to the data warehouse, as did over 3,500 vendor partners. The system handled up to 35,000 queries per week, with benefits per query estimated to be over $12,000 per query. Power users ran up to 1,000 queries per day.

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[38] Foote, K. P. S. and Rishnamurthi, M. (2001). Forecasting using data warehousing model: Wal-Mart's experience. *The Journal of Business Forecasting*, 13–17.

10.5.8 Caveats

Data mining involves the application of statistics and artificial intelligence. In business, one of the most popular forms of data mining is support to customer relationship management (CRM). While CRM is very promising, it has often been found to be less effective than hoped. Patton (2001)[39] found that up to 70 percent of CRM projects did not produce measurable business benefits. CRM systems can cost up to $70 million to develop, with additional expenses incurred during implementation. Patton cited problems with applications at Monster.com, Mshow, and CopperCom. One reason cited for problems in implementing CRM was that its users, marketing personnel for the most part, were not as computer-familiar as were accounting and production personnel. At Mshow, the sales force refused to use a new CRM system. At CopperCom, a $500,000 CRM project was reduced in size due to lack of support from an applications service provider. On the other hand, Siebel Systems, the largest provider of CRM software, reported that the vast majority of its customers were happy with their product. Even Fingerhut, a pioneer at using data mining for business, has seen their operations shut down after being absorbed by a large sales organization.

Many of the problems in CRM expectations have been blamed on over zealous sales pitches. CRM offers a lot of opportunities to operate more efficiently. However, they are not silver bullets, and benefits are not unlimited. As with any system, prior evaluation of benefits is very difficult, and investment in CRM systems needs to be based on sound analysis and judgment.

10.6 ONLINE ANALYTICAL PROCESSING

Another type of application to exploit data warehouses might not be as primitive in terms of the analysis conducted, but is tremendously fast in response and enables executives to make timely decisions: online analytical processing (OLAP). Tables, even if joining data from several sources, limit the review of information. Often, executives need to view information in multiple combinations of two dimensions. For example, an executive might want to see a summary of the quantity of each product sold in each region. Then, she might want to view the total quantities of each product sold within each city of a region. In addition, she might want to view quantities sold of a specific product in all cities of all regions. OLAP is specially designed to answer queries such as these. OLAP applications let a user rotate virtual "cubes" of information, whereby each side of the cube provides another two dimensions of relevant information.

[39] Patton, S. (2001). The truth about CRM. *CIO Magazine*, www.cio.com/archive/050101/truth_content.html.

10.7 QUESTIONS AND ANSWERS RELEVANT TO BI

1. I have heard of Competitive Intelligence. How is Business Intelligence different?

Competitive Intelligence is one aspect of Business Intelligence. Competitive Intelligence is limited to information about competitors and how that affects strategy, tactics and operations. Business Intelligence includes that, but also uses information about overall industry conditions, political situations, and internal operations.

2. How does Business Intelligence relate to Knowledge Management?

We define Knowledge Management as the way that organizations create, capture and re-use knowledge to achieve organizational objectives. There's some overlap between the two terms, but there seem to be some differences of application in most cases. Business Intelligence uses many of the tools of Knowledge Management (things like data mining, for example) but concentrates more on strategic and tactical recommendations than Knowledge Management.

3. What are the three key steps in Business Intelligence?

The three key steps in Business Intelligence are gathering, analysis and recommendation. For most organizations, this is most effective if it's a part of regular ongoing business practice.

4. Should Business Intelligence be a staff function, or should I use an outsider?

The answer to that is an absolute "it depends." Larger businesses often have staff resources that can do most of the data gathering and analysis. They will often benefit though from having an outside perspective during the analysis phase. Smaller businesses almost never have the internal staff resources to do an effective Business Intelligence job. The exception to this is if there is a person (often the CEO) who does the BI work because he or she likes it. Smaller businesses and independent business professionals can usually benefit from outside help with information gathering and with analysis and recommendation.

5. Who is Business Intelligence for?

Business Intelligence is for businesses that want to drive intelligent decision-making throughout their organizations and make it easy for everyone in the organization to collaborate, analyze, share, and act on business information from a centrally managed, more secure source. With its attractive price for the Enterprise model, Business Intelligence supports IT professionals, information workers, and developers, and empowers organizations of all sizes.

10.8 ALL ABOUT ITeS

As per the encyclopedia.thefreedictionary.com, ITeS stands for "Information technology enabled services, or ITeS, a form of outsourced service which has emerged due to involvement of IT in various fields such as banking and finance, telecommunications, insurance, etc. Some of the examples of ITeS are medical transcription, back-office accounting, insurance claim, credit card processing and many more." IT Enabled Services are human-concentrated services that are transported over telecommunication networks or the Internet to a variety of business segment. ITeS significantly augments the employment opportunities. According to Carl Gunnarsen, "ITeS stands for IT-enabled services that major corporations

are now adapting into their businesses." For a company to be considered ITeS-powered, it should have:

1. The kind of services that are being offered through the use of telecommunication. It can also be through networking and other forms of electronic media.
2. Services that make use of information technology from the process up to the delivery of the product.

ITeS makes possible the following:

- Outsourcing of processes that can be enabled with information technology.
- Delivered from/to remote areas through the telecom and Internet medium.
- Transfer of ownership and management of the process from the customer to the service provider, e.g., functions like finance, HR (human resource), administration, healthcare, telecommunication, engineering etc.

Figure 10.20 shows the uses of ITeS.

Some ITeS are:

- e-Governance.
- Call Centers.
- Data Management.
- Medical [Telemedicine and Transcription].
- Data Digitization.
- Website Services.

ITeS are applied to many areas of work including:

- Program Management.
- Enterprise IT Policy and Planning.
- Enterprise Design.

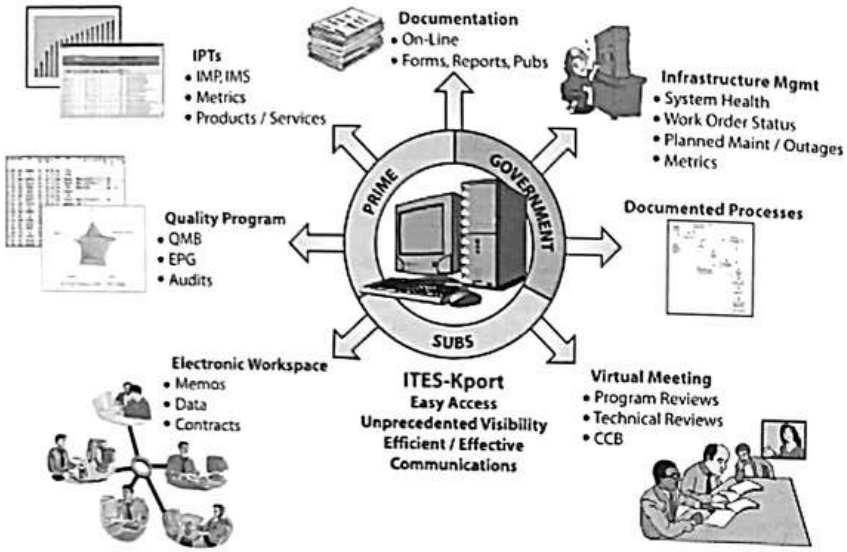


Figure 10.20: Diagrammatical Explanation of ITeS.
Source: Tim Rouch, Northrop Grumman Corporation (2005).[40]

- Integration and Consolidation.
- Information Assurance.
- Business Process Re-engineering.
- Requirements Analysis.
- Market Research and Prototyping.
- Information and Knowledge Engineering.
- Development of Software Interfaces and Software Configuration.
- Product Integration.
- Test and Evaluation.
- IT Infrastructure Systems Integration.
- Seat Management/Asset Management & Managed Services.
- Technology Insertion.

[40] Northrop Grumman Corporation (Northrop Grummer) is engaged into manufacturing of products, services, acrospace, electronics and shipbuilding services.

In order to operate ITeS, the following basic skills are required:

- Basic Computer knowledge (for general processes).
- Excellent Communication Skills (neutral Accent, clear speech, good grammar & language etc.).
- Customer Interaction Skills (customer handling skills, telephone etiquette, empathy and listening skills etc.).
- Understanding of target groups and their cultures.
- Positive attitudes like:
  - ➢ commitment to understanding customers.
  - ➢ solving their problems.
  - ➢ motivation to work hard.
- Process training is given in-house by ITeS companies but basic skills like communication and interaction skills etc. are expected attributes of the employment candidate and form the criteria of screening for ITeS jobs.

Table 10.8 lists skills required by functional activity.

There are many areas promising high growth in the use of ITeS:

- Customer Analytics and CRM.
- HR Outsourcing.
- Legal transcript support.
- Knowledge process outsourcing —
  - ➢ Market research.
  - ➢ Equity research.
  - ➢ Engineering design.
  - ➢ Animation and simulation.
  - ➢ R & D (non-IT areas).
- Financial Services outsourcing ... already witnessing explosive growth —
  - ➢ Insurance underwriting.
  - ➢ Fund management.

Table 10.8: Skills Requirements by Function.

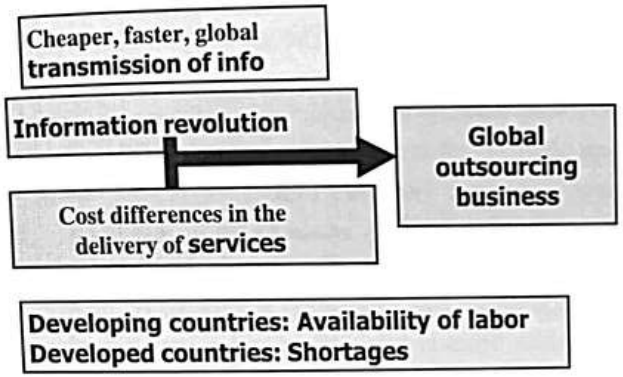
Function	Skills required
Call center	Good communication and language skills, accent understanding team leadership, basic computing skills
Remote customer interaction	Language and accent understanding
Date search, Integration	Computing, language and analytical skills
Human Resource Services	Country-specific HR policies, rules and regulations
Remote education	Subject knowledge, computing and language skills
Engineering and design	Technical and engineering design and computing skills
Translation, medical transcription and Localization	Language understanding, basic computing (word processing knowledge) and understanding of various medical terminologies
Animation	Drawing and creative skills, computer graphic skills
Finance and accounting	International/country-specific accounting rules
Market Research	Understanding statistical sales and marketing concepts
Network Consultancy and management	Understanding different network configurations and support equipment, technical/computing skills

- Risk assessment and actuarial analytics.
- Commodities processing services.
- Debt collection and recovery.
- Equity research.
- Financial data mining and modeling.
- Corporate & Market research.

ITeS were designed to help customers meet enterprise standards at every level while meeting local customer requirements. ITeS puts control of the contractor in the hands of the local customer who can continue to work with one contractor as a program matures or changes scope. (ITeS becomes your vehicle.) ITeS can be used for almost anything

from hardware to services. Figure 10.21 shows some of the driving forces behind ITeS.

Table 10.9 lists companies who are providing services under the ITeS banner.



NIRMA INSTITUTE OF MANAGEMENT Library AHMEDABAD

Figure 10.21: Driving Forces behind ITeS.

Table 10.9: ITeS Providers.

Accenture National Security Services	HP
Advanced Systems Technology	Linden International
American Systems Corporation	Microsoft Corporation
Applied Information Sciences, Inc.	MORI Associates, Inc.
Base Technologies, Inc.	Multimax
Capstone	NETCONN Solutions
CEXEC, Inc.	Optimus Corporation
CISCO Systems, Inc.	PAVL
Communications Technologies, Inc.	PKMM
Data Systems Analysts, Inc.	RAM
Dell	Securify, Inc.
DeVA Systems, Inc.	Smart Technology, Inc.
DigitalNet	Solid Network Solutions
EDO	Strategic Enterprise Systems, Inc.
EDS	Techni-Core Engineering
Federated Data	TelosOK
Frontier	Titan
Hoppmann	T-Systems

10.9 CONCLUSION

During the last 10 to 15 years, business management has changed intensely around the world. Business is increasingly becoming *digital*. In the future, even more than today, most business will be either digital or depend critically on aspects that are digital. The global economy is increasingly a *knowledge* economy. Products and services will be developed and delivered, competition won or lost, and jobs defined, by knowledge — its creation, organization, and communication. Information technologies are pervasive, and transform what they touch.

The invention of technology made Business Intelligence (BI) possible over relational engines, but now the experiences of linking them with information technology has unearthed a new set of problems. Business intelligence enables firms to take informed business decisions. It involves assembling, accumulating, analyzing, and accessing corporate data. For this purpose, it uses a variety of tools including query and reporting tools, OLAP, DM and DSS. Business intelligence is increasingly being identified as a mission-critical, firm-wide technology which helps firms to create business advantage and reap the rewards of expensive applications such as CRM and SCM (Supply Chain Management).

This chapter has introduced the concepts of data warehousing and business intelligence as enhancements to EIS systems. EIS systems and business intelligence operations are mutually supportive. Both can exist without the other, but both can be much more profitable if used together. Data warehouses are often used as efficient storage units for larger EIS systems. They also provide the ability to support business intelligence in the form of data mining. Data mining has proven to be extremely effective in improving many business operations. The process of data mining relies heavily on information technology, in the form of data storage support (data warehouses, data marts, and/or online analytic processing tools) as well as software to analyze the data (data mining software). However, the process of data mining is far more than simply applying these data mining software tools to a firm's data. Intelligence is required on the part of the analyst in selection of model types, in selection and transformation of the data relating to the specific problem, and in interpreting results.

By going through this lesson, you are now able to:

- Know how technical components of Business Intelligence (software, a wide variety of technical equipment) are managed in theory.
- Develop an ability to transfer and relate the formal knowledge of theory to practice.
- Initiate and execute independently various types of projects, many of which are IT-related.
- Understand business intelligence as learning in action.
- Understand modern information technology and its business opportunities.

You are thus equipped with the valuable, sought-after skills that will give you an advantage as a prospective employee.

☆ KEYWORDS

- ☆ **Analytical Model:** A structure and process for analyzing a dataset. For example, a decision tree is a model for the classification of a dataset.
- ☆ **Business Intelligence (BI):** The activity of gathering information about the elements in the environment that interacts with your firm.
- ☆ **Churn:** Customer turnover to competitor promotional programs.
- ☆ **Customer Relationship Management (CRM):** Software with the purpose of monitoring customer requirements and aiding organizations in maintaining their business. Data mining support of CRM includes identification of the value of a customer profile.
- ☆ **Data Integrity:** Absence of meaningless, corrupt, or redundant data.
- ☆ **Data Mart:** Either smaller scale data warehouse, or database product gathering data for a specific study.
- ☆ **Data Mining:** Use of statistics or artificial intelligence to confirm hypotheses or to identify patterns.

- ☆ **Data Quality:** Accuracy and completeness of data with proper formatting.
- ☆ **Data Redundancy:** The same data being shuffled from one place to the next, leading to massive amounts of data being repeated and taking up significant amounts of storage and processing power.
- ☆ **Data Standardization:** Process of developing unique variable values.
- ☆ **Data Warehouse:** The data warehouse is the place where transactional source data are specifically structured for query and analysis performance and ease of use. It is also known as subject-oriented, integrated, time-variant, and non-volatile collection of data.
- ☆ **Granular data:** Degree of detail in data.
- ☆ **Hypothesis Testing:** Systematic statistical testing of a proposed relationship.
- ☆ **Integrated data:** The data from various subject areas should be rationalized with one another.
- ☆ **Knowledge Discovery:** Examining data without preconceived notions of relationships.
- ☆ **Matching:** Association of variables within a database.
- ☆ **Micromarketing:** Targeting promotional campaigns based on information about customer's likelihood of response.
- ☆ **Online Analytical Processing (OLAP):** Refers to array-oriented database applications that allow users to view, navigate, manipulate, and analyze multi-dimensional databases.
- ☆ **Scalable:** Ability of the system to cope with increases in volume.
- ☆ **Subject-oriented:** Data is organized around a major object or process of an organization. Classic examples include subject area databases for customer, material, vendor, and transaction.
- ☆ **Time-variant:** Record is accurate only as at a specific moment in time. In some cases, the moment in time is a single moment. In other cases, it is a span of time. In any case, the values of data found in a data warehouse are accurate and relevant only to a specific moment in time.
- ☆ **Versatility:** Ability of the system to apply a wide variety of data mining models.

☆ TERMINOLOGY REVIEW

1. Discuss the role of BI in business today.
2. Discuss the role of BI and DM in the various fields of management, such as marketing, human resource, finance, and production.
3. Explain the purpose of an (info cube/cube) and dimensions. How do two components relate to each other? Why is it important to identify an organization's key performance indicators before developing a BI solution?
4. The Business Intelligence solution has been designed to interoperate with data that exists in virtually any enterprise data source, Explain.
5. How does BI work with existing servers and databases?
6. What are the system requirements for Business Intelligence?
7. How does Business Intelligence fit in with the Operating system?
8. How much does Business Intelligence cost?
9. What is Business Intelligence and what does it do?
10. What if I have a small (fewer than 100-person) company? Can I still use Business Intelligence?
11. What is a typical way an organization might use Business Intelligence?
12. What programs are included in Business Intelligence? Is Business Intelligence available as a single product in a box?
13. Describe a data warehouse. What does that have to do with EIS?
14. What is data granularity?
15. What are data marts typically used for in CRM?
16. What is an On-Line Analytic Processing System, and what relation does it have with data mining?
17. What is meant by data quality?
18. What is meant by data integrity?
19. What is meant by data standardization?
20. Describe micromarketing, and its relationship with data mining.
21. What is the difference between hypothesis testing and knowledge discovery?
22. What does scalable mean in information systems?
23. What is the difference between IT and ITeS?

24. What does ITeS mean?
25. Which types of work are available in ITeS industry?
26. What do I need to enter the ITeS arena?
27. What is meant by churn in data mining?
28. How does an organization get intelligence from cross-pollination of business information? The lists below provide an example of the questions that result from the hybrid mix of customer and competitor information.

☆ COMPETENCY REVIEW

A. Student Activity

1. To run a company with just traditional BI tools would be like driving a car and looking only into the rear-view mirror. Although you can see everything that happens, you don't see it until it has happened, which might be too late. Comment.
2. Business Intelligence includes two major components: the business intelligence platform and end-user tools. Justify your answer with illustrations.
3. Experience is about creating something that people want to use. People are happier with a EIS software product when they enjoy using it. For instance, Ted refers to Tableau as "a radically new product." I've seen it and it's a GREAT experience, with some GREAT visualization but there's nothing REVOLUTIONARY about it except for the experience. It's not in the cloud, it's not scaling beyond the petabytes, it's not even a web product (it's a windows desktop APP). Not revolutionary, just GREAT to use.
4. Search the library and/or the Internet for applications of CRM in EIS.
5. Search the library and/or the Internet for applications of data warehouses in EIS.
6. What happens when software becomes a commodity? There's usually a mid-market but you start to see players emerge at two ends of a spectrum.

7. Business intelligence is arguably the most important enterprise application in your company. But questions abound: How can you get more workers using it? What are the new frontiers? And how simple can you make it?

B. Explain the Following Statements with Relevance to Businesses Intelligence

1. The good side of the 80/20 rule still applies. Experience-based doesn't always mean 100% high end, every bell and whistle.
2. Focus on features that matter to the user doing a job. If a feature is needed to help a customer nail a part of their using your product, add it and make it better than they expect. Lacking features isn't a bad thing if you keep adding them — for instance, the iPhone was deficient in features (no GPS, battery was a pain, etc.) but users were patient.
3. Provide a high quality product that is as much about using as performing. The experienced-based product says that it's not enough to have a product that does what you want, but it has to be something you ENJOY using.
4. User and Experience is KING. Usability is not something that is a feature to implement, it's the thing that informs, prioritizes and determines what features are implemented.

C. Briefly Comment on the Following

a. The business intelligence that results from cross-pollination establishes reason and basis, the knowledge for actions.

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b. A business is operated with the objective of making a profit from the sale of goods or services.

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c. Business Intelligence enables comprehension, understanding and profit from experience.

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d. Business intelligence is not a single entity; it is decomposed into business information.

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